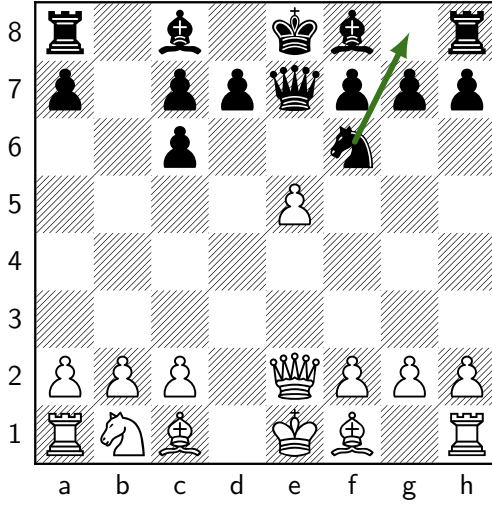


Melanie-MASTERS (human masters) — Scotch Game

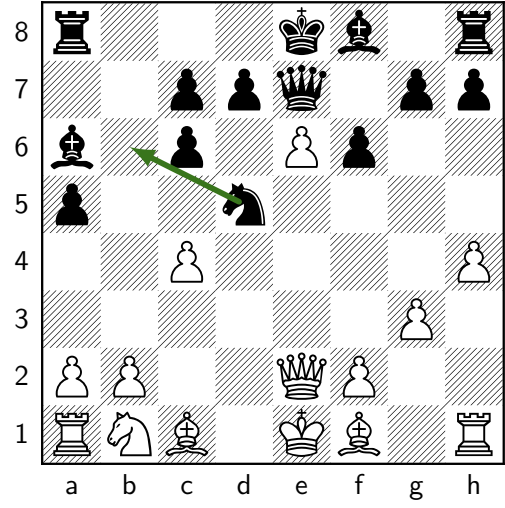
Melanie (Deep_Vision)

Abstract

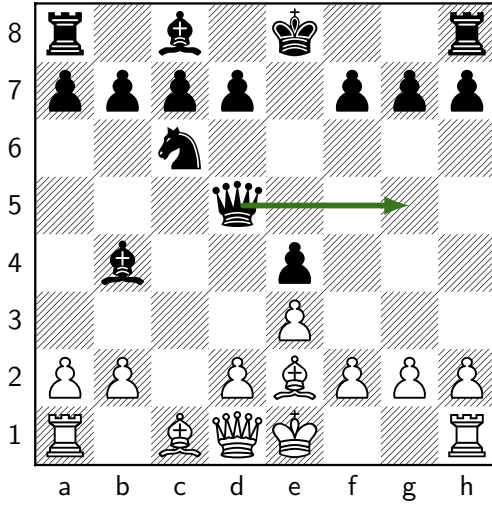
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



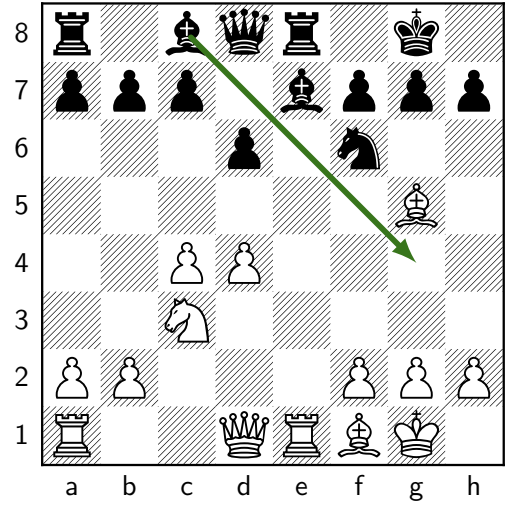
(a)



(b)

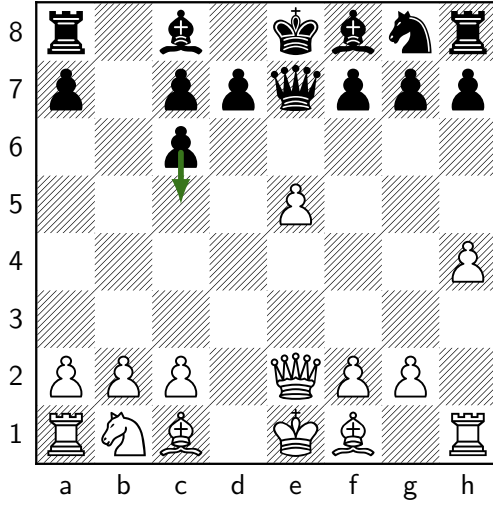


(c)

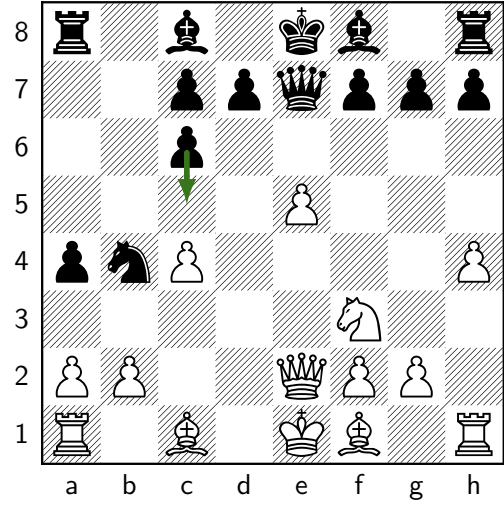


(d)

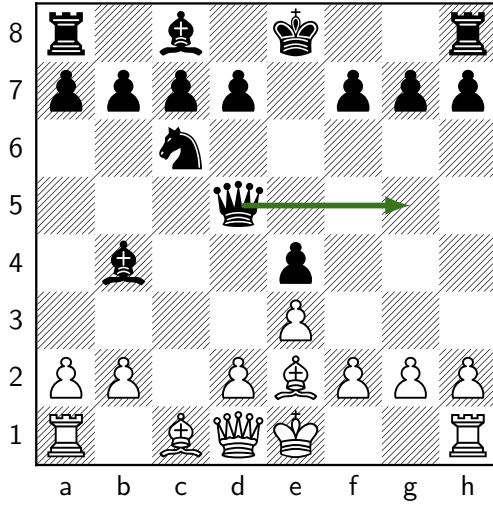
Figure 1: **Move 7, Black: Ng8.** (a) Our position; Melanie plays Ng8 (f6g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#148` (ply 21) [structural distance $d=79$, QAP obj=74]: the master played Nb6; under μ our Nf3 plays the role of that game’s Nd4; cross-type roles: our Bc8 as their Rh8, our Nf6 as their Pd7; transferred move f6g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#145` (ply 21) [structural distance $d=78$, QAP obj=37]: the master played Qg5; under μ our Qe2 plays the role of that game’s Qd4; cross-type roles: our Bc8 as their Ra8, our Nf6 as their Bc8; transferred move e7b4. (d) Retrieved from `lichess_elite_2024-09_p0013#2791` (ply 25) [structural distance $d=80$, QAP obj=37]: the master played Bg4; under μ our Pc3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Ra8, our Pd7 as their Qd8; transferred move c6c5. Synthesis: 1 retrieved analog transferred to Ng8 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



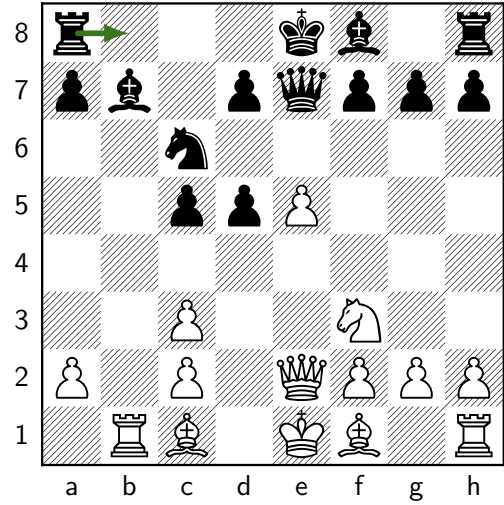
(a)



(b)

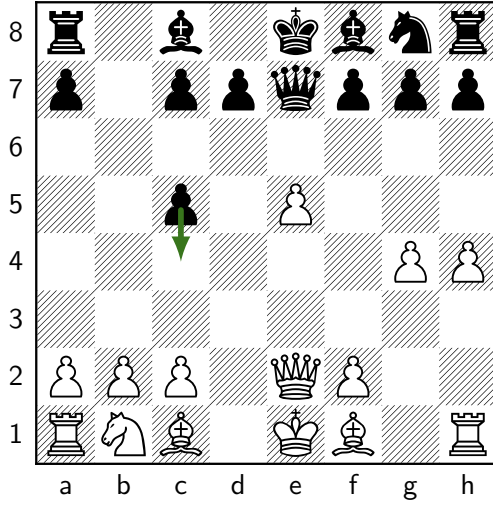


(c)

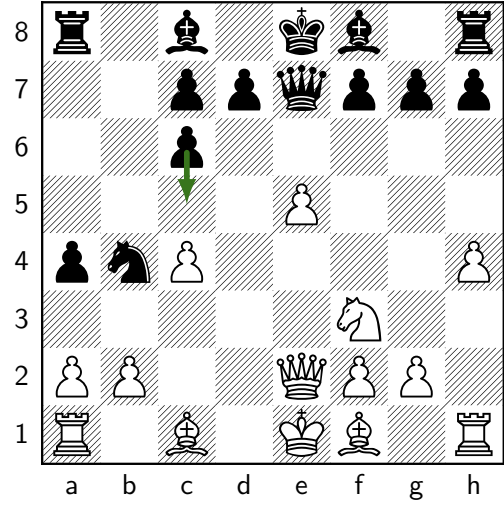


(d)

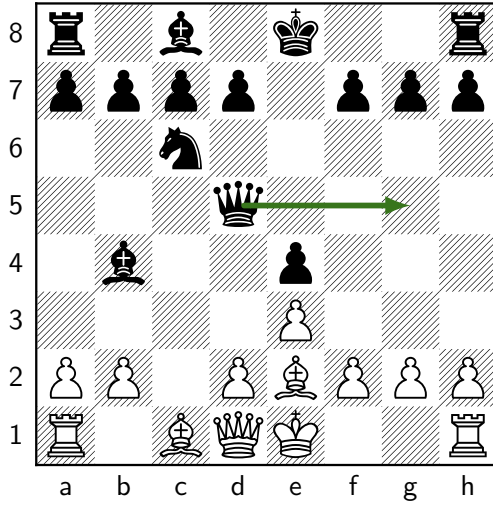
Figure 2: **Move 8, Black: c5.** (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#127` (ply 21) [structural distance $d=77$, QAP obj=104]: the master played c5; under μ our *Pc3* plays the role of that game’s *Pc3*; cross-type roles: our *Ng8* as their *Rh8*, our *Qe7* as their *Nb4*; transferred move c6c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#145` (ply 21) [structural distance $d=72$, QAP obj=95]: the master played Qg5; under μ our *Qe2* plays the role of that game’s *Qd4*; cross-type roles: our *pf2* as their *Rh8*, our *Ng8* as their *Pa7*; transferred move e7b4. (d) Retrieved from `lichess_elite_2024-09_p0022#3258` (ply 21) [structural distance $d=71$, QAP obj=86]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Ng8* as their *Bf8*, our *pe5* as their *Bb7*; transferred move a8b8. Synthesis: 1 retrieved analog transferred to c5 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



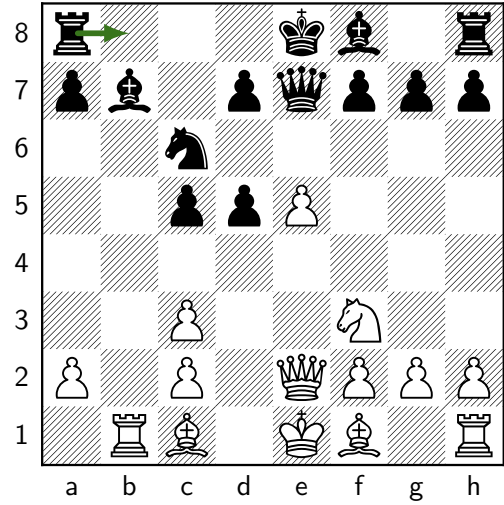
(a)



(b)

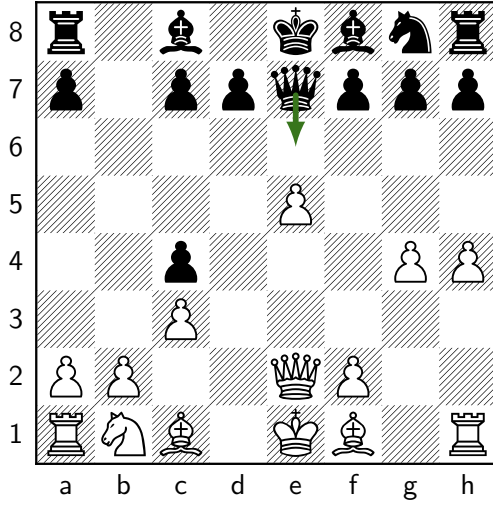


(c)

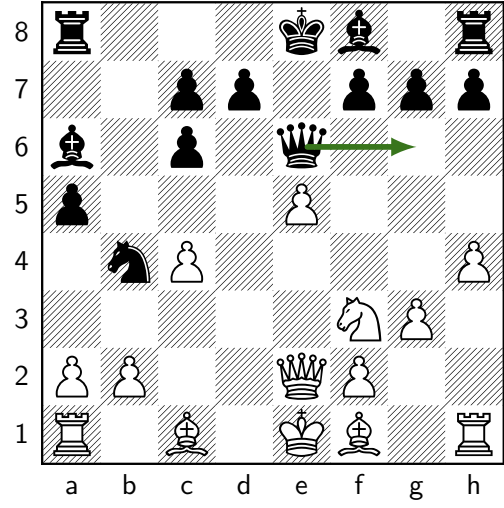


(d)

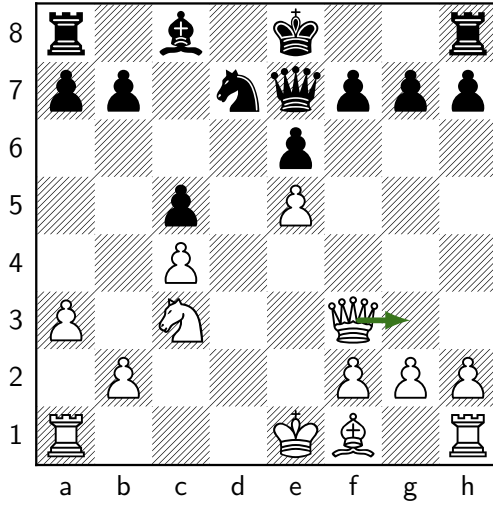
Figure 3: **Move 9, Black: c4.** (a) Our position; Melanie plays c4 (c5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#127` (ply 21) [structural distance $d=77$, QAP obj=104]: the master played c5; under μ our *Pc4* plays the role of that game’s *Pc3*; cross-type roles: our *Bc8* as their *Ra8*, our *Pc7* as their *Bc8*; transferred move c5c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#145` (ply 21) [structural distance $d=87$, QAP obj=95]: the master played Qg5; under μ our *Qe2* plays the role of that game’s *Qd4*; cross-type roles: our *Bf8* as their *Ra8*, our *pa2* as their *Bc8*; transferred move e7h4. (d) Retrieved from `lichess_elite_2024-09_p0022#3258` (ply 21) [structural distance $d=68$, QAP obj=86]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Bc8* as their *Rh8*, our *pe5* as their *Bb7*; transferred move a8b8. Synthesis: 1 retrieved analog transferred to c4 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



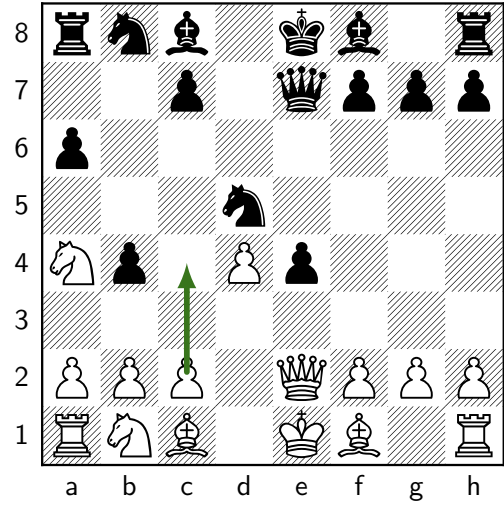
(a)



(b)

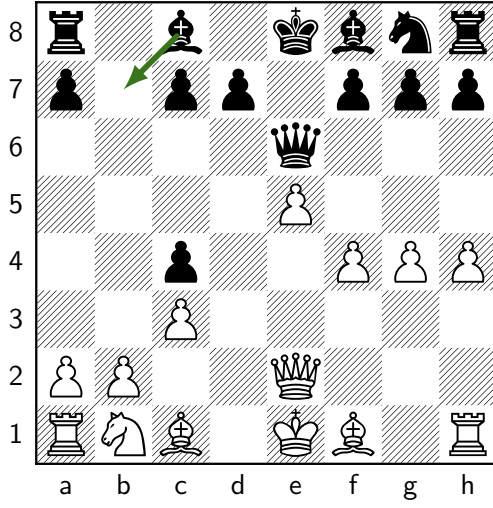


(c)

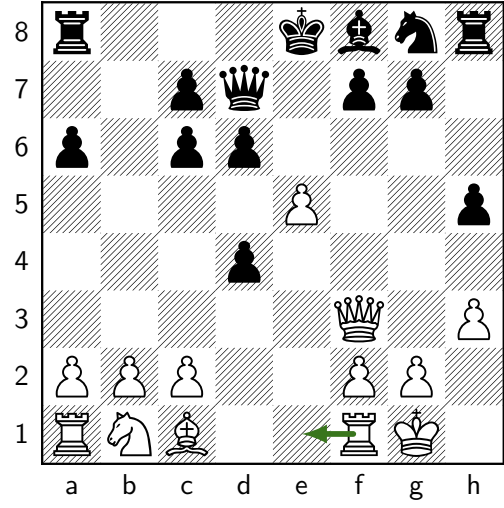


(d)

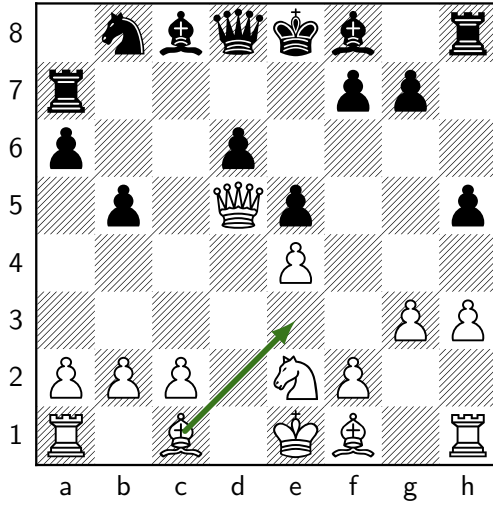
Figure 4: **Move 10, Black: Qe6.** (a) Our position; Melanie plays Qe6 (e7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3690` (ply 23) [structural distance $d=80$, QAP obj=64]: the master played Qg6; under μ our Qe2 plays the role of that game’s Qe3; cross-type roles: our Ng8 as their Ra8, our Pc7 as their Bf8; transferred move e7e6 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#1910` (ply 22) [structural distance $d=88$, QAP obj=62]: the master played Qg3; under μ our Qe2 plays the role of that game’s Qf3; cross-type roles: our Bc8 as their Ra1, our Pd7 as their Bf1; transferred move e7e6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#592` (ply 26) [structural distance $d=73$, QAP obj=87]: the master played c4; under μ our Pd2 plays the role of that game’s Pc2; cross-type roles: our Ng8 as their Ra1, our Pc4 as their Bc1; transferred move d7d5. Synthesis: 2 retrieved analogs transferred to Qe6 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



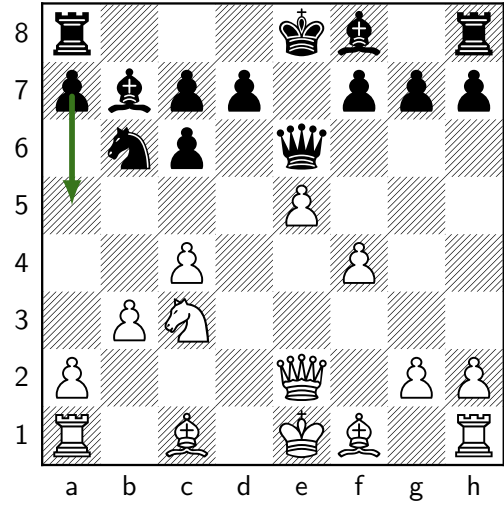
(a)



(b)

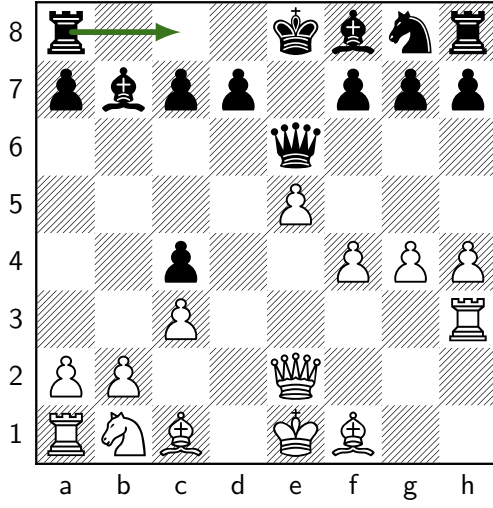


(c)

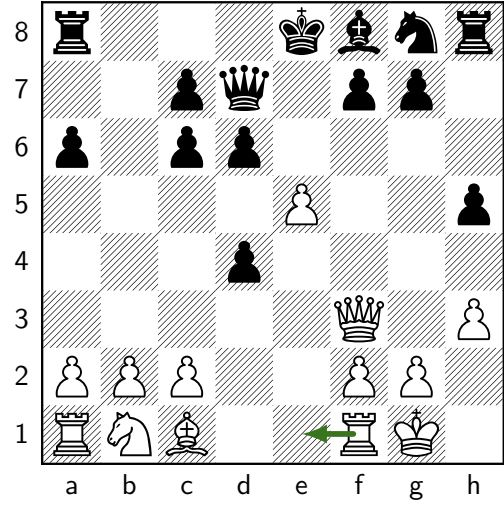


(d)

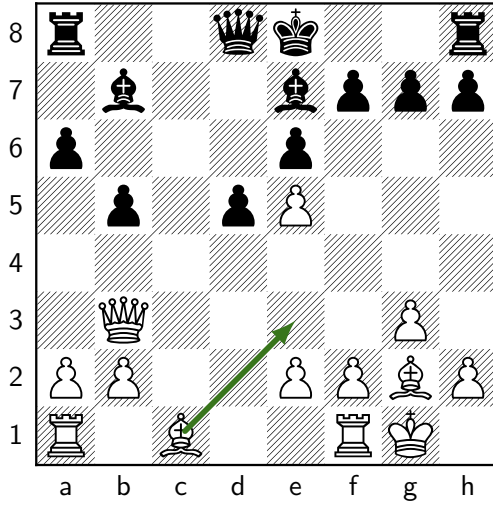
Figure 5: **Move 11, Black: Bb7**. (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2329` (ply 20) [structural distance $d=69$, QAP obj=78]: the master played R $e1$; under μ our Bc1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Nb1, our ph4 as their Bc1; transferred move c8b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3591` (ply 20) [structural distance $d=66$, QAP obj=127]: the master played Be3; under μ our Ph2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Ra1, our pa2 as their Bf1; transferred move h7h5. (d) Retrieved from `lichess_elite_2024-09_p0022#2522` (ply 21) [structural distance $d=70$, QAP obj=99]: the master played a5; under μ our Pa2 plays the role of that game’s Pa2; cross-type roles: our Bc8 as their Rh8, our Pc4 as their Bb7; transferred move a7a5. Synthesis: 1 retrieved analog transferred to Bb7 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



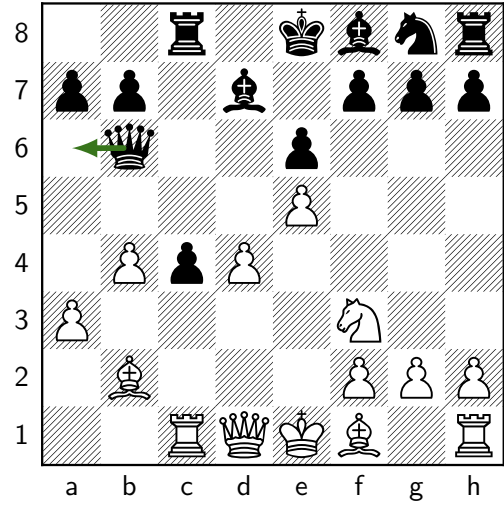
(a)



(b)

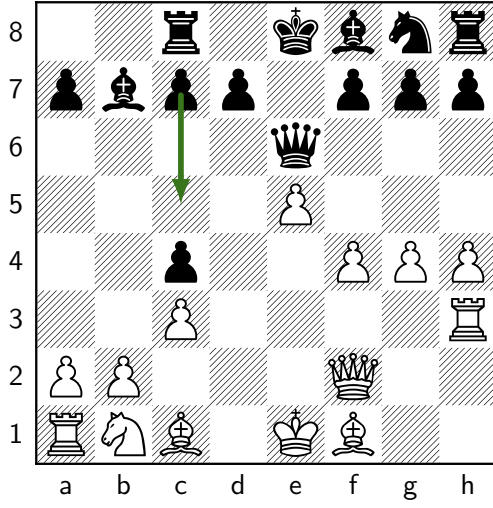


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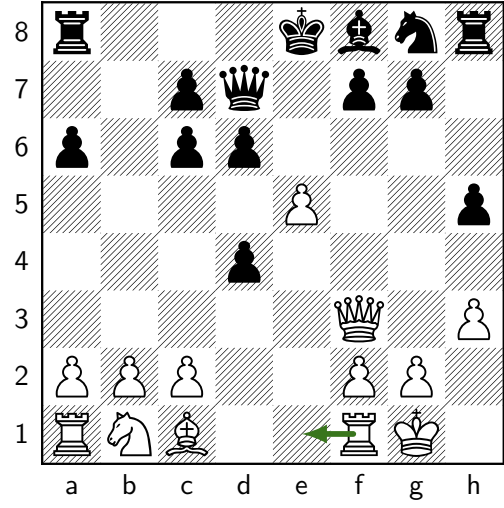


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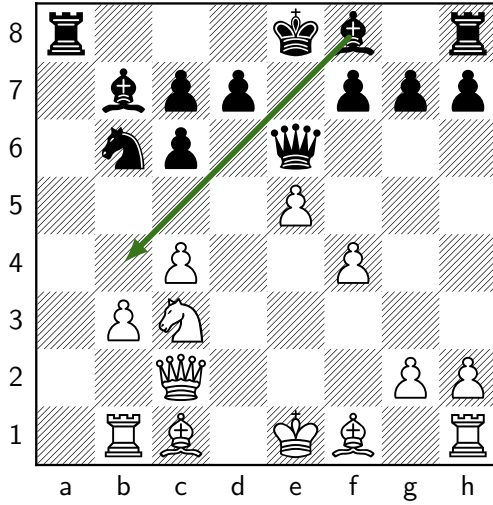
Figure 6: **Move 12, Black: Rc8**. (a) Our position; Melanie plays Rc8 (a8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2329` (ply 20) [structural distance $d=67$, QAP obj=99]: the master played R $e1$; under μ our R $a1$ plays the role of that game’s R $f1$; cross-type roles: our p $a2$ as their R $a1$, our R $h8$ as their N $b1$; transferred move a8b8. (c) Retrieved from `twic1500#186` (ply 26) [structural distance $d=83$, QAP obj=67]: the master played B $e3$; under μ our P $a2$ plays the role of that game’s B $c1$ (a **cross-type** correspondence); cross-type roles: our B $f8$ as their R $a1$, our p $a2$ as their R $f1$; transferred move a7a6. (d) Retrieved from `lichess_elite_2024-09_p0022#1925` (ply 23) [structural distance $d=69$, QAP obj=63]: the master played Q $a6$; under μ our Q $e3$ plays the role of that game’s Q $b3$; cross-type roles: our Q $e6$ as their B $f8$, our P $a7$ as their N $g8$; transferred move e6a6. Synthesis: no retrieved analog transferred to Rc8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



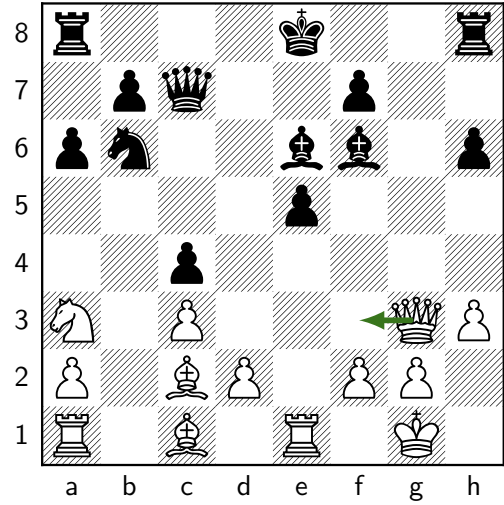
(a)



(b)

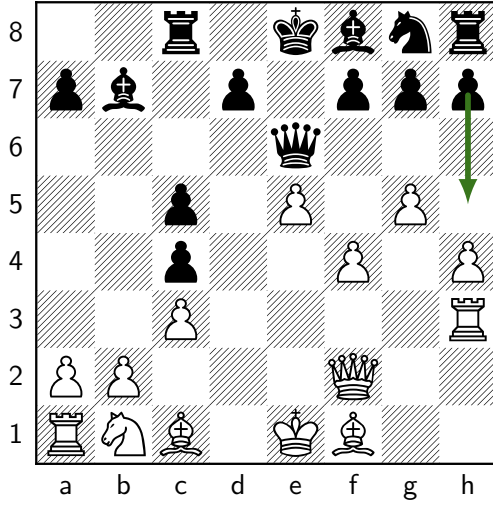


(c)

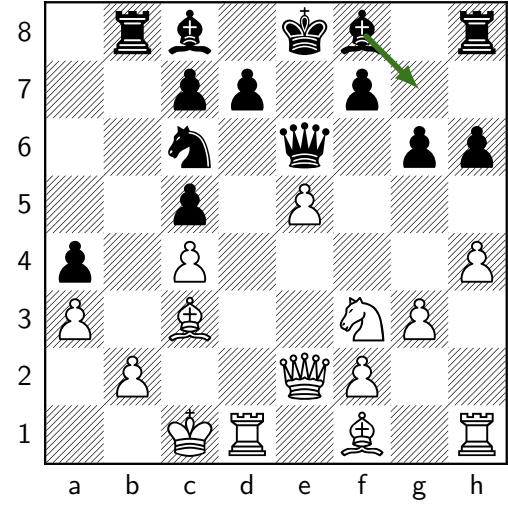


(d)

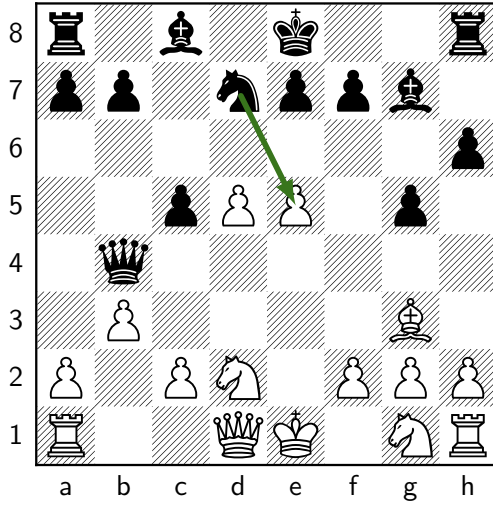
Figure 7: **Move 13, Black: c5.** (a) Our position; Melanie plays c5 (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2329` (ply 20) [structural distance $d=52$, QAP obj=94]: the master played Re1; under μ our Pc2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Ra1, our Rc8 as their Nb1; transferred move c7c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#2522` (ply 27) [structural distance $d=77$, QAP obj=95]: the master played Bb4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Ng8 as their Pc7, our Qe6 as their Nb6; transferred move f8b4. (d) Retrieved from `lichess_elite_2025-11_p0000#1141` (ply 36) [structural distance $d=77$, QAP obj=47]: the master played Qf3; under μ our Qe3 plays the role of that game’s Qg3; cross-type roles: our Ng8 as their Ra1, our Pf7 as their Bc1; transferred move e6d5. Synthesis: 1 retrieved analog transferred to c5 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



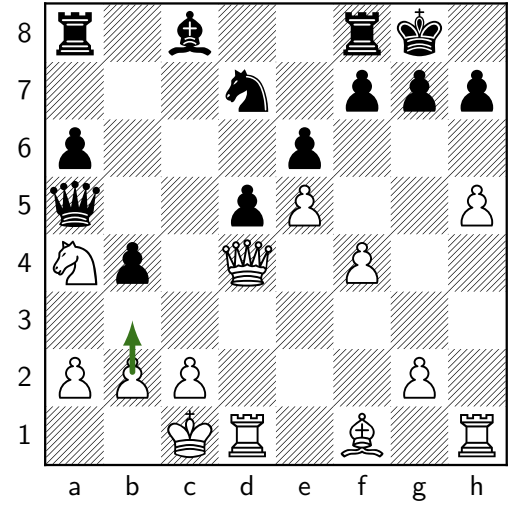
(a)



(b)

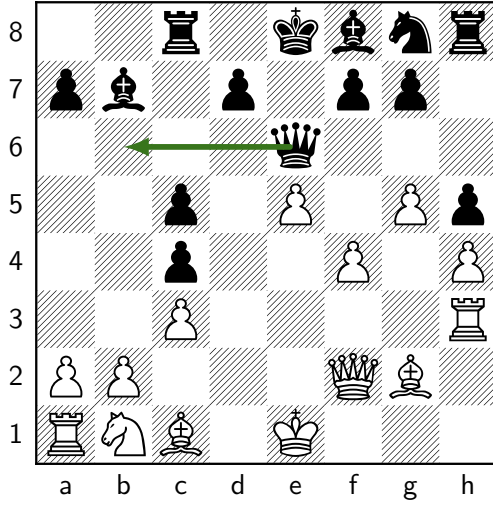


(c)

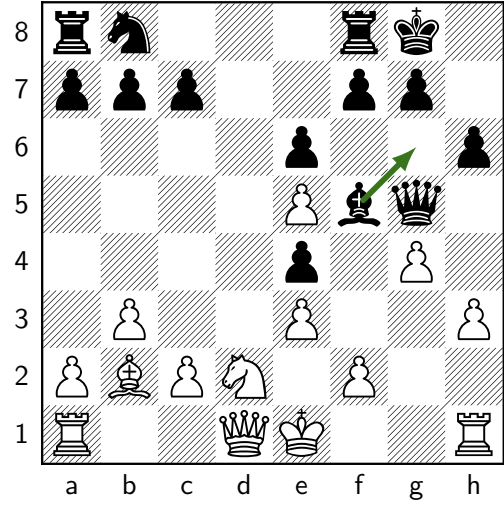


(d)

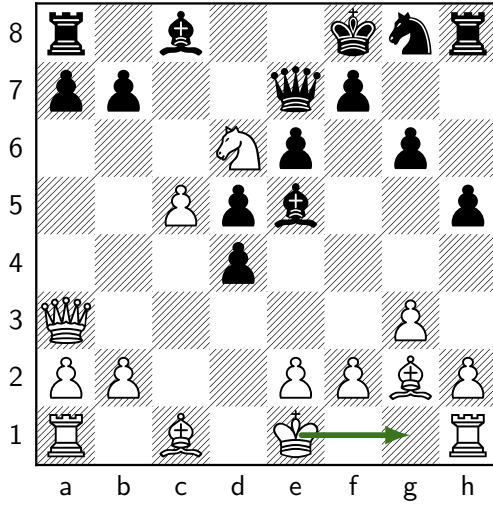
Figure 8: **Move 14, Black: h5.** (a) Our position; Melanie plays h5 (h7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#127` (ply 33) [structural distance $d=81$, QAP obj=59]: the master played Bg7; under μ our *Ph2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pg7* as their *Rb8*, our *Rc8* as their *Bc8*; transferred move h7h5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1257` (ply 27) [structural distance $d=91$, QAP obj=46]: the master played Nxe5; under μ our *Qe3* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Rh8*, our *Rh8* as their *Pa7*; transferred move e6d6. (d) Retrieved from `lichess_elite_2024-09_p0013#3093` (ply 30) [structural distance $d=92$, QAP obj=40]: the master played b3; under μ our *Bb2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Rd1*, our *Pd7* as their *Bf1*; transferred move b7a8. Synthesis: 1 retrieved analog transferred to h5 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



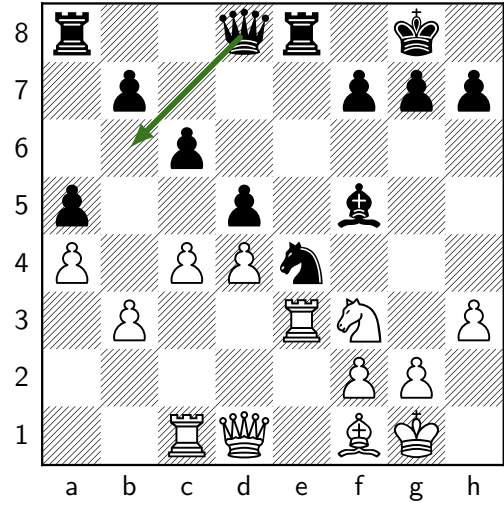
(a)



(b)

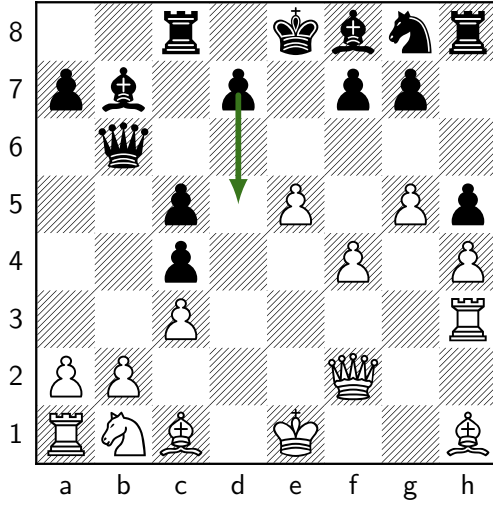


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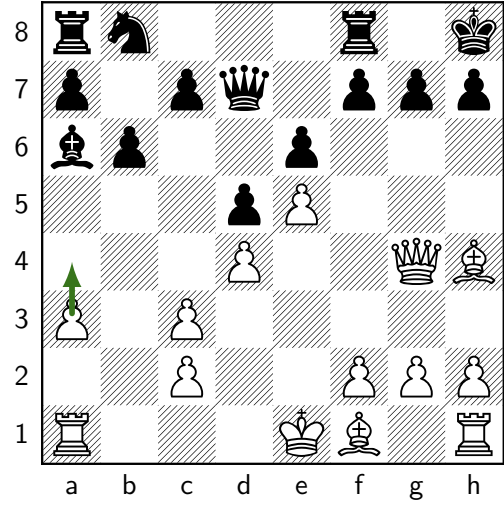


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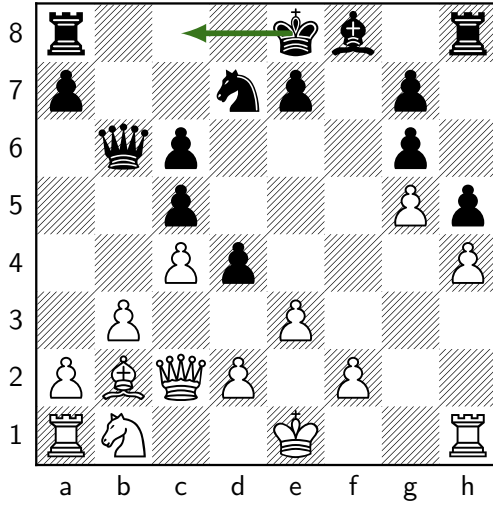
Figure 9: **Move 15, Black: Qb6.** (a) Our position; Melanie plays Qb6 (e6b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#2804` (ply 23) [structural distance $d=91$, QAP obj=30]: the master played Bg6; under μ our Qe3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Rf8, our Ng8 as their Pa7; transferred move e6b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#154` (ply 24) [structural distance $d=97$, QAP obj=20]: the master played 0-0; under μ our Ke1 plays the role of that game’s Ke1; cross-type roles: our bc1 as their Ra1, our Pa7 as their Bc1; transferred move e8d8. (d) Retrieved from `lichess_elite_2025-11_p0000#1519` (ply 39) [structural distance $d=103$, QAP obj=20]: the master played Qb6; under μ our Rh1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Ra8, our Bf8 as their Pf7; transferred move h8h7. Synthesis: 1 retrieved analog transferred to Qb6 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



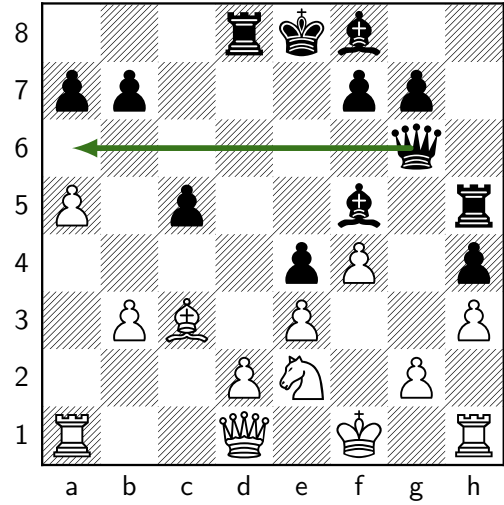
(a)



(b)

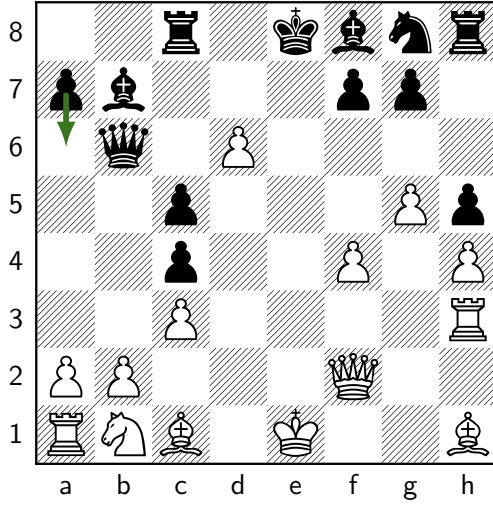


(c)

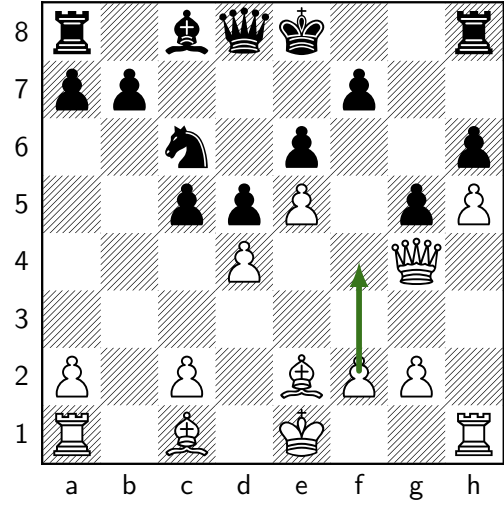


(d)

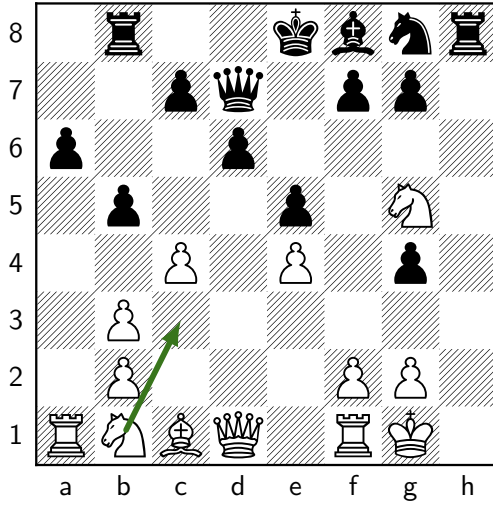
Figure 10: **Move 16, Black: d5**. (a) Our position; Melanie plays d5 (d7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#1842` (ply 24) [structural distance $d=87$, QAP obj=66]: the master played a4; under μ our *Pd2* plays the role of that game’s *Pa3*; cross-type roles: our *Qb6* as their *Ph2*, our *ph4* as their *Bh4*; transferred move d7d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2312` (ply 27) [structural distance $d=106$, QAP obj=66]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ng8* as their *Ra8*, our *Rh8* as their *Bf8*; transferred move e8e7. (d) Retrieved from `lichess_elite_2024-09_p0013#737` (ply 37) [structural distance $d=97$, QAP obj=53]: the master played Qa6; under μ our *Qb3* plays the role of that game’s *Qg3*; cross-type roles: our *gf2* as their *Rd8*, our *Pc4* as their *Rh5*; transferred move b6a6. Synthesis: 1 retrieved analog transferred to d5 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



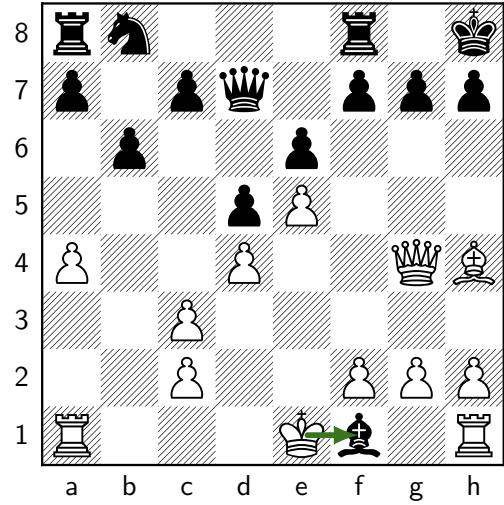
(a)



(b)

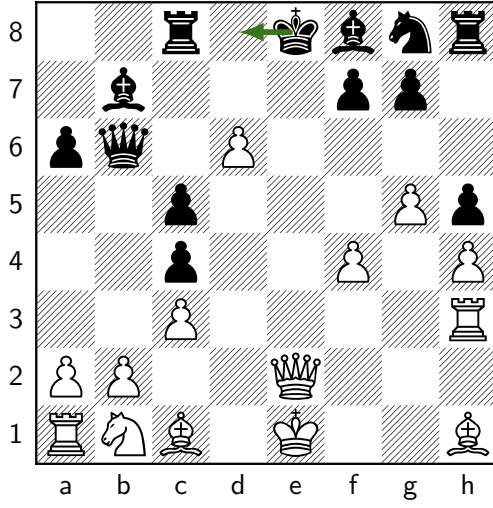


(c)

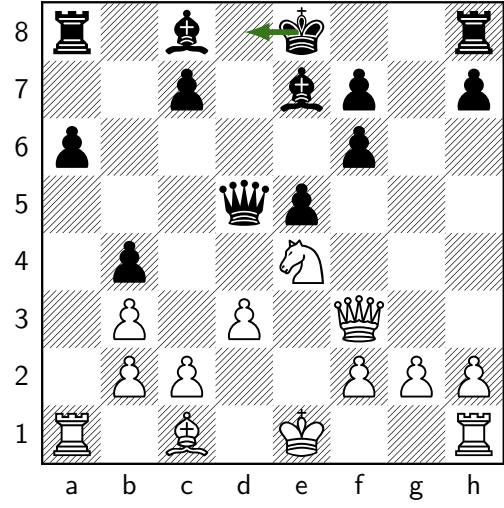


(d)

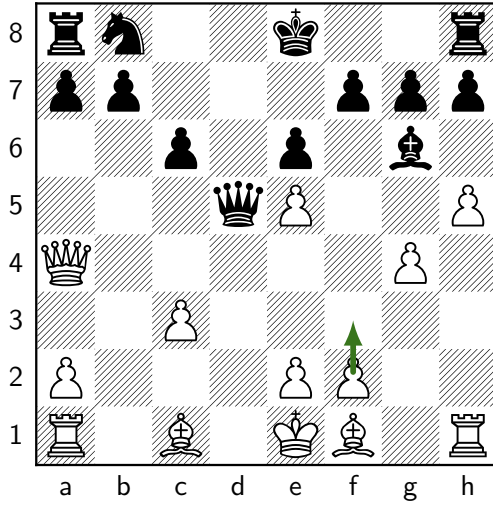
Figure 11: **Move 17, Black: a6** (a7a6). (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#10` (ply 26) [structural distance $d=78$, QAP obj=74]: the master played f4; under μ our *Pa2* plays the role of that game’s *Pf2*; cross-type roles: our *Ph5* as their *Ra1*, our *Rh8* as their *Pc2*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2188` (ply 24) [structural distance $d=76$, QAP obj=52]: the master played Nc3; under μ our *Rh1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *pg5* as their *Bc1*, our *pc3* as their *Rf1*; transferred move h8h7. (d) Retrieved from `lichess_elite_2024-09_p0030#1842` (ply 26) [structural distance $d=81$, QAP obj=20]: the master played Kxf1; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pg5* as their *Ra1*, our *qf2* as their *bf1*; transferred move e8d8. Synthesis: 1 retrieved analog transferred to a6 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



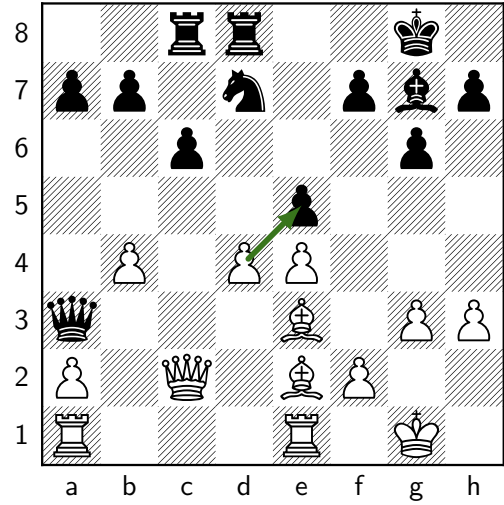
(a)



(b)

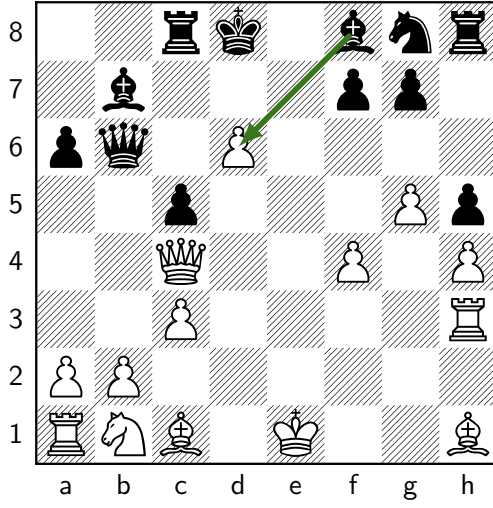


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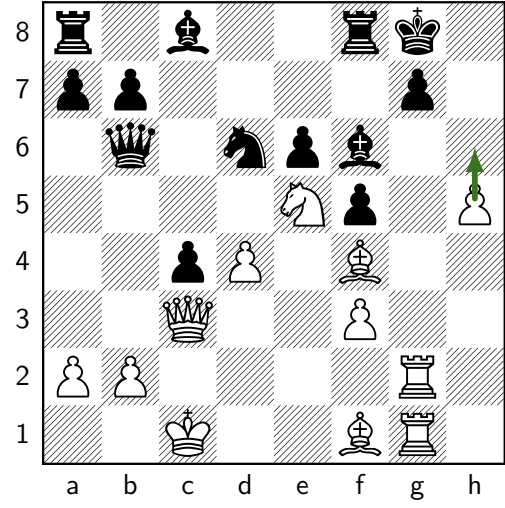


(d)

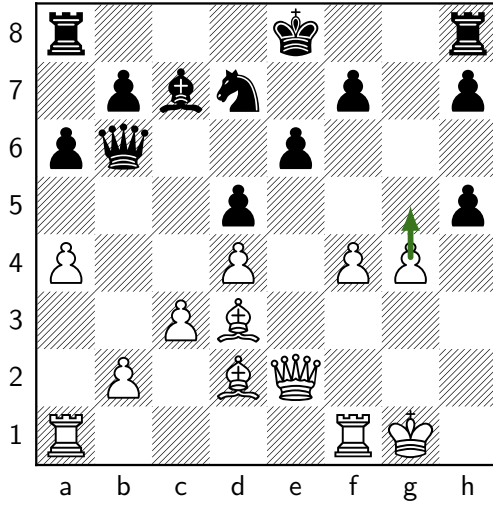
Figure 12: **Move 18, Black: Kd8.** (a) Our position; Melanie plays Kd8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1403` (ply 27) [structural distance $d=102$, QAP obj=5]: the master played Kd8; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pf7* as their *Ra8*, our *ra1* as their *Bc8*; transferred move e8d8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3595` (ply 26) [structural distance $d=89$, QAP obj=28]: the master played f3; under μ our *Bb2* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *pg5* as their *Bc1*, our *Ph5* as their *Bf1*; transferred move b7e4. (d) Retrieved from `lichess_elite_2024-09_p0018#3792` (ply 38) [structural distance $d=94$, QAP obj=16]: the master played dxe5; under μ our *Bb2* plays the role of that game’s *Pd4* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Ra1*, our *pc3* as their *Be2*; transferred move b7e4. Synthesis: 1 retrieved analog transferred to Kd8 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



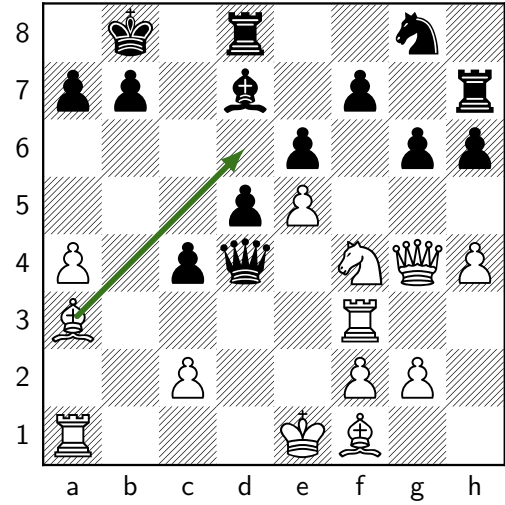
(a)



(b)

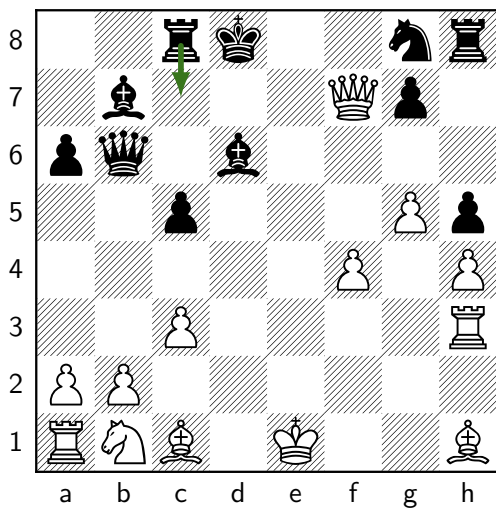


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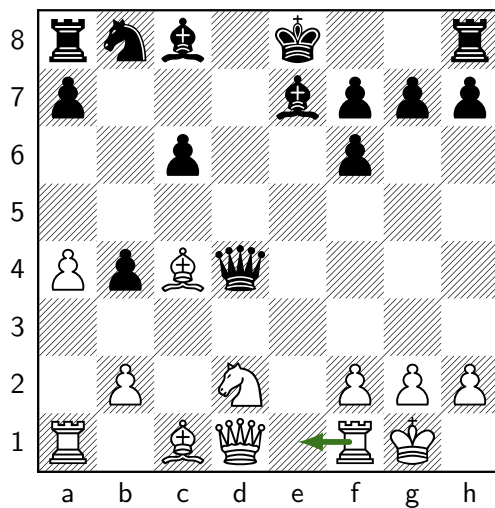


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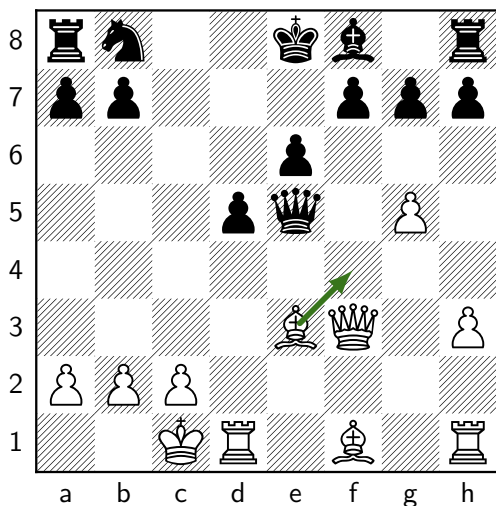
Figure 13: **Move 19, Black: Bxd6.** (a) Our position; Melanie plays Bxd6 (f8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#3089` (ply 44) [structural distance $d=99$, QAP obj=40]: the master played h6; under μ our *Bf1* plays the role of that game’s *Ph5* (a **cross-type** correspondence); cross-type roles: our *pc3* as their *Bf1*, our *nb1* as their *Rg1*; transferred move f8d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3106` (ply 38) [structural distance $d=108$, QAP obj=72]: the master played g5; under μ our *Bb2* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *pc3* as their *Ra1*, our *ph4* as their *Bd2*; transferred move b7a8. (d) Retrieved from `lichess_elite_2024-09_p0031#3758` (ply 38) [structural distance $d=106$, QAP obj=60]: the master played Bd6+; under μ our *Bb2* plays the role of that game’s *Ba3*; cross-type roles: our *Pc5* as their *Ra1*, our *Ph5* as their *Bf1*; transferred move b7g2. Synthesis: 2 retrieved analogs transferred to Bxd6 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



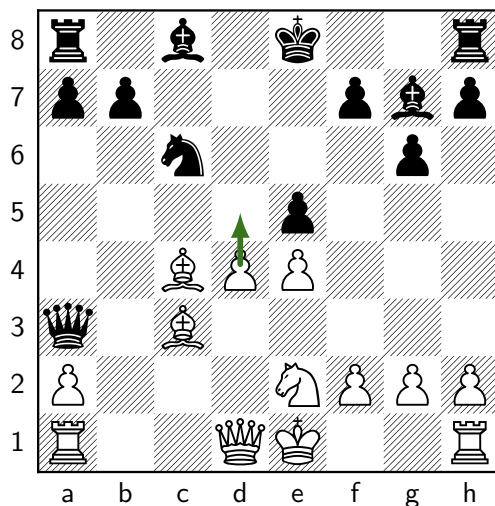
(a)



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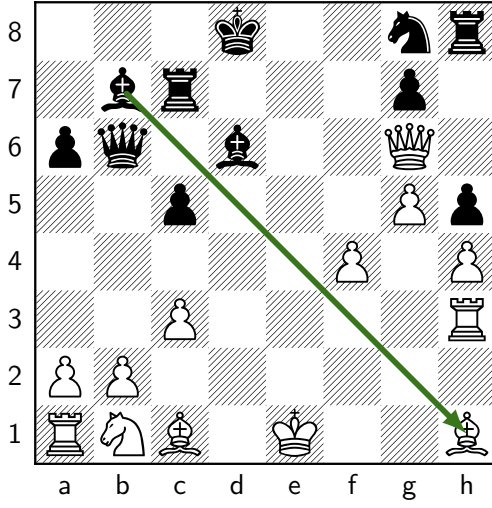


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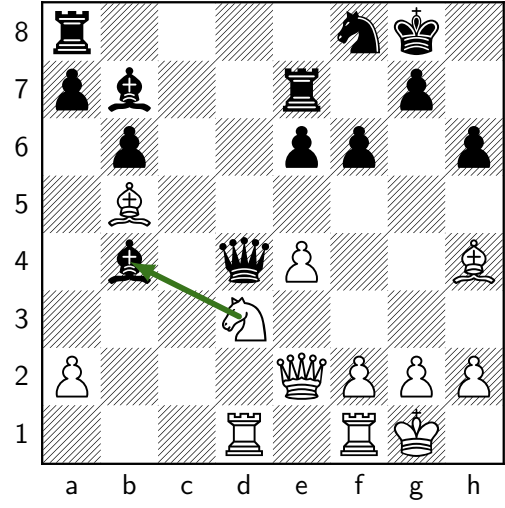


(d)

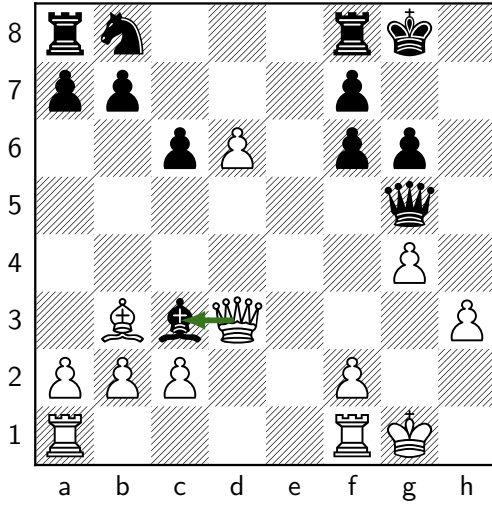
Figure 14: **Move 20, Black: Rc7**. (a) Our position; Melanie plays Rc7 (c8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#392` (ply 24) [structural distance $d=112$, QAP obj=42]: the master played Re1; under μ our Rc1 plays the role of that game’s Rf1; cross-type roles: our Qb6 as their Ra1, our nb1 as their Bc1; transferred move c8b8. (c) Retrieved from `lichess_elite_2024-09_p0031#3483` (ply 24) [structural distance $d=89$, QAP obj=42]: the master played Bf4; under μ our Ng1 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Rh1, our Bd6 as their Pc2; transferred move g8f6. (d) Retrieved from `lichess_elite_2024-09_p0020#3201` (ply 24) [structural distance $d=112$, QAP obj=26]: the master played d5; under μ our Bb2 plays the role of that game’s Pd4 (a **cross-type** correspondence); cross-type roles: our Rc8 as their Qd1, our Qb6 as their Rh1; transferred move b7e4. Synthesis: no retrieved analog transferred to Rc7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



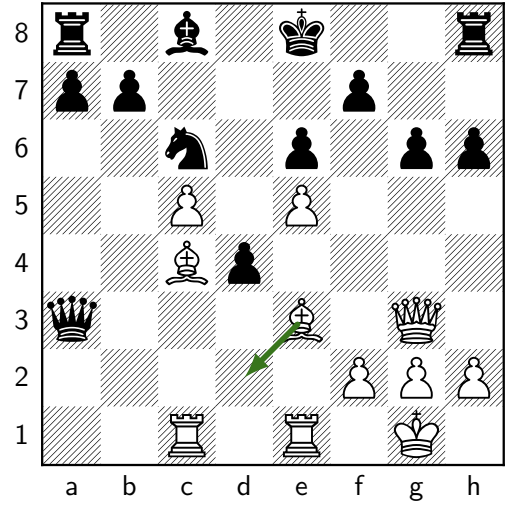
(a)



(b)

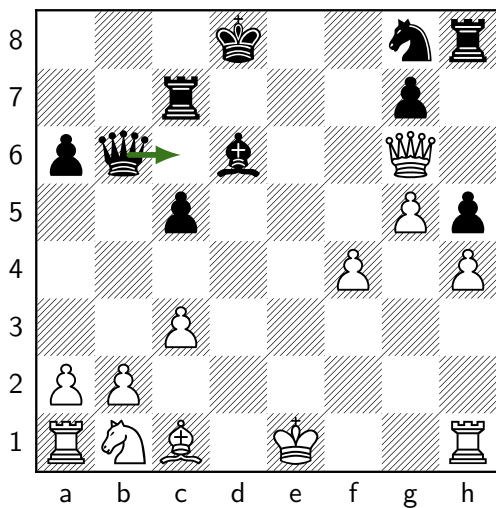


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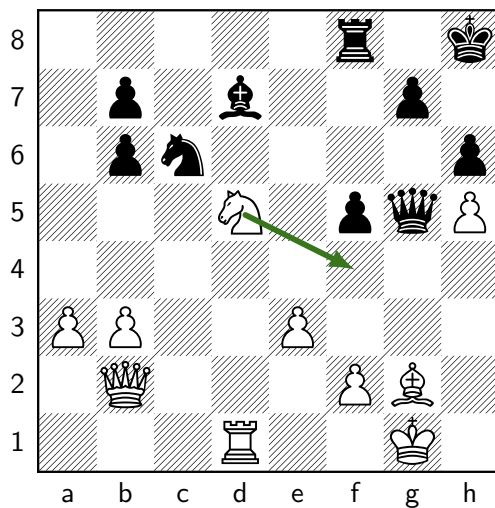


(d)

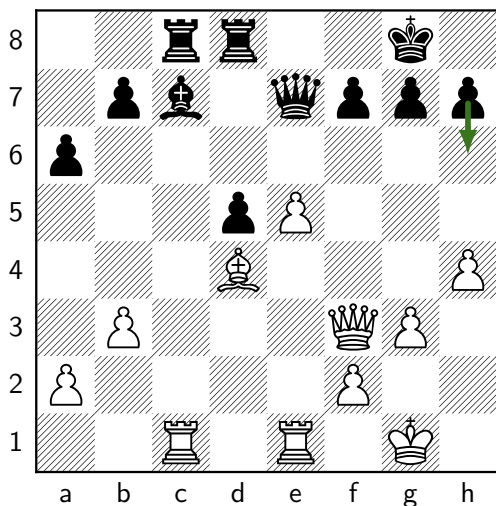
Figure 15: **Move 21, Black: Bxh1.** (a) Our position; Melanie plays Bxh1 (b7h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2379` (ply 46) [structural distance $d=105$, QAP obj=56]: the master played Nx4; under μ our Bd3 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our pb2 as their Rd1, our Qb6 as their Rf1; transferred move d6f8. (c) Retrieved from `lichess_elite_2024-09_p0016#3601` (ply 34) [structural distance $d=113$, QAP obj=55]: the master played Qxc3; under μ our Rh1 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Ng8 as their Ra1, our Bd6 as their Pb2; transferred move h8h7. (d) Retrieved from `lichess_elite_2024-09_p0010#138` (ply 36) [structural distance $d=97$, QAP obj=55]: the master played Bd2; under μ our Bd3 plays the role of that game’s Be3; cross-type roles: our Pg7 as their Rc1, our pg5 as their Bc4; transferred move d6f8. Synthesis: no retrieved analog transferred to Bxh1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



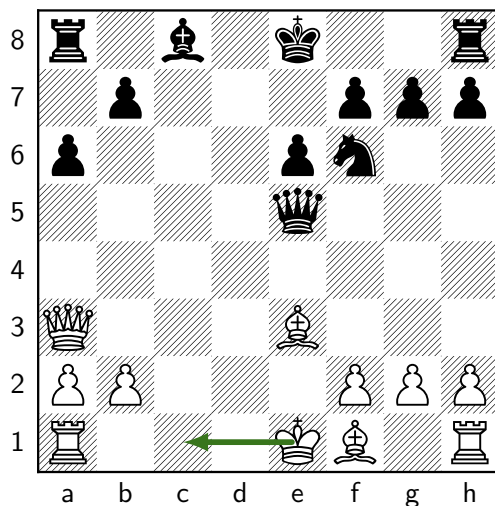
(a)



(b)

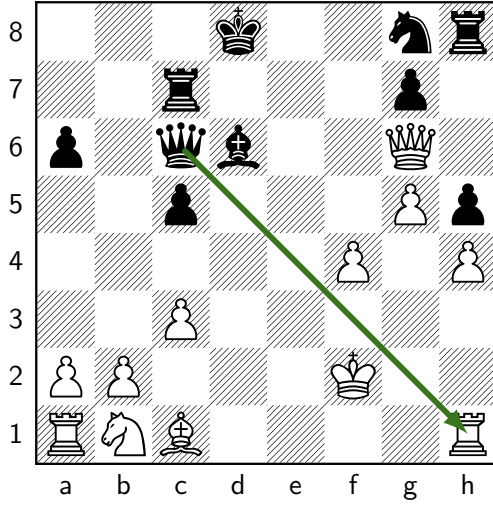


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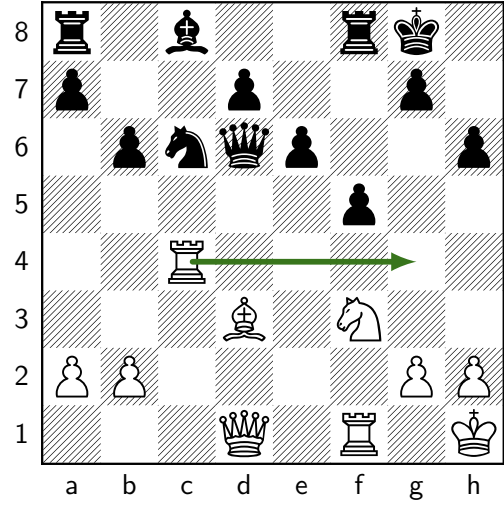


(d)

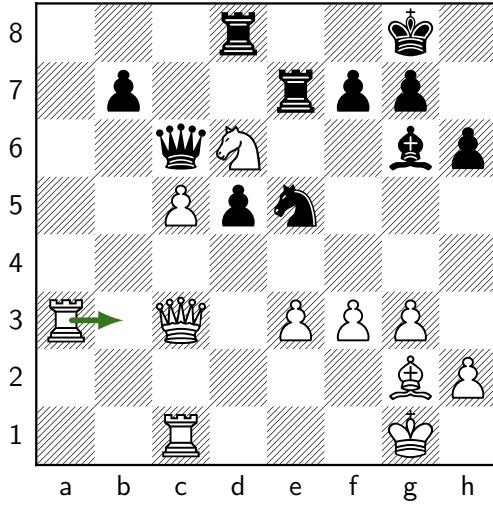
Figure 16: **Move 22, Black: Qc6**. (a) Our position; Melanie plays Qc6 (b6c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#290` (ply 56) [structural distance $d=93$, QAP obj=89]: the master played Nf4; under μ our Qb3 plays the role of that game’s Nd5 (a **cross-type** correspondence); cross-type roles: our ph4 as their Qb2, our Ng8 as their Pf2; transferred move b6c6 (**endorses this move**). (c) Retrieved from `lichess_elite.2024-09_p0019#3371` (ply 43) [structural distance $d=80$, QAP obj=70]: the master played h6; under μ our Pc4 plays the role of that game’s Ph2; cross-type roles: our Qb6 as their Rc8, our Rh8 as their Pb7; transferred move c5c4. (d) Retrieved from `lichess_elite.2024-09_p0031#3812` (ply 26) [structural distance $d=92$, QAP obj=66]: the master played O-O-O; under μ our Kd1 plays the role of that game’s Ke1; cross-type roles: our Pc5 as their Ra1, our Ng8 as their Bf1; transferred move d8d7. Synthesis: 1 retrieved analog transferred to Qc6 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



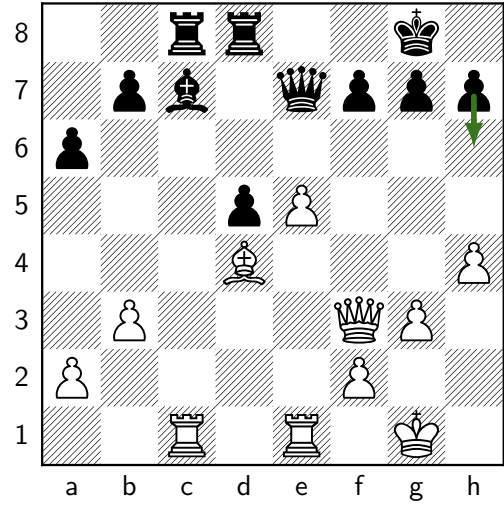
(a)



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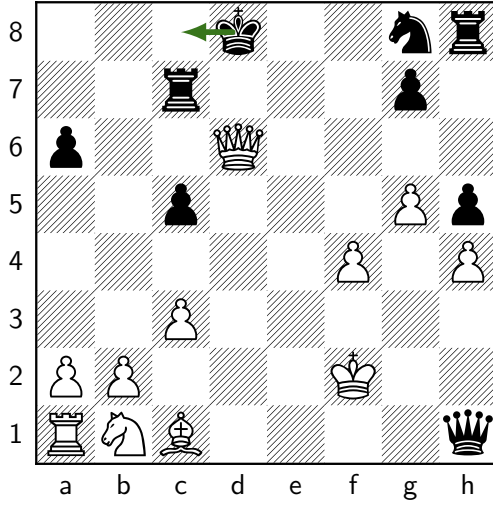


(c)

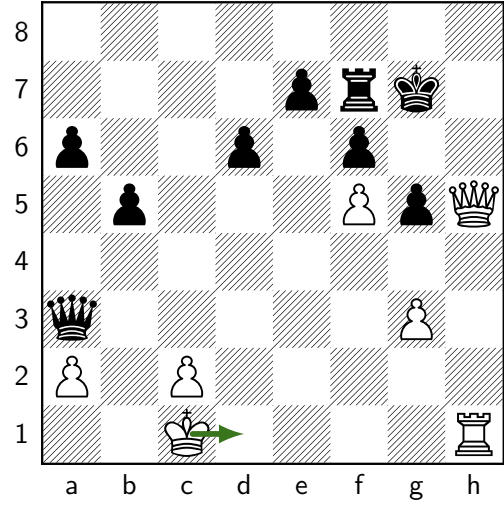


(d)

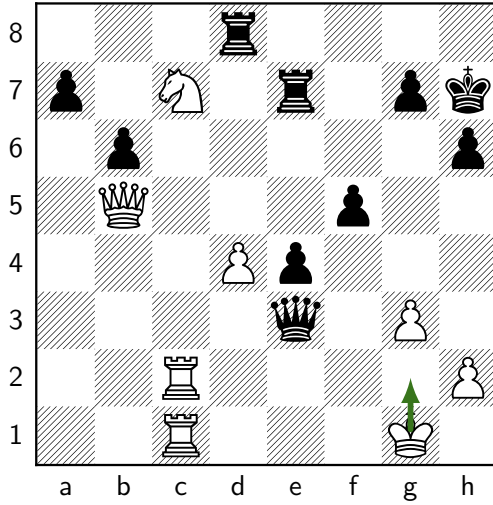
Figure 17: **Move 23, Black: Qxh1.** (a) Our position; Melanie plays Qxh1 (c6h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1170` (ply 38) [structural distance $d=92$, QAP obj=92]: the master played Rg4; under μ our Qc3 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rf1, our Ng8 as their Pa2; transferred move c6b6. (c) Retrieved from `lichess_elite_2024-09_p0030#1170` (ply 56) [structural distance $d=88$, QAP obj=74]: the master played Rb3; under μ our Pc4 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Rh8 as their Bg2, our Ng8 as their Pe3; transferred move c5c4. (d) Retrieved from `lichess_elite_2024-09_p0019#3371` (ply 43) [structural distance $d=93$, QAP obj=71]: the master played h6; under μ our Rh1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Qc6 as their Rc8, our Ng8 as their Pb7; transferred move h8h7. Synthesis: no retrieved analog transferred to Qxh1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



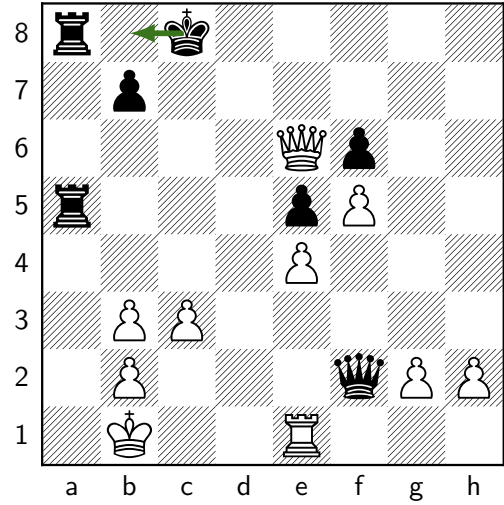
(a)



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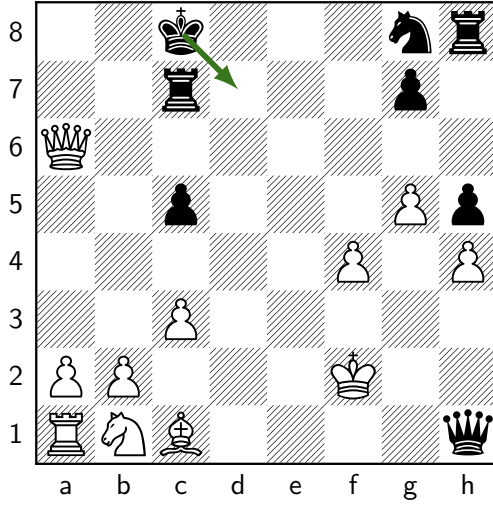


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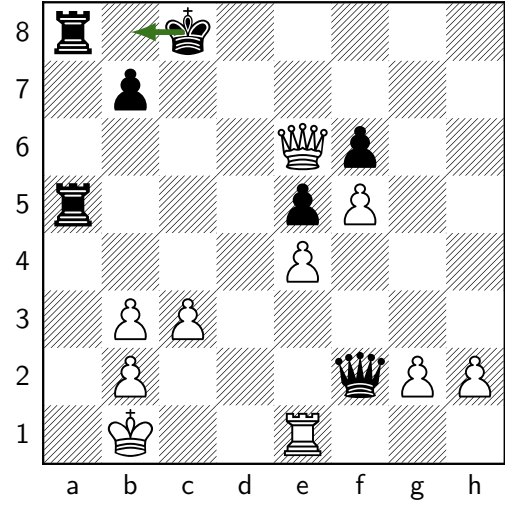


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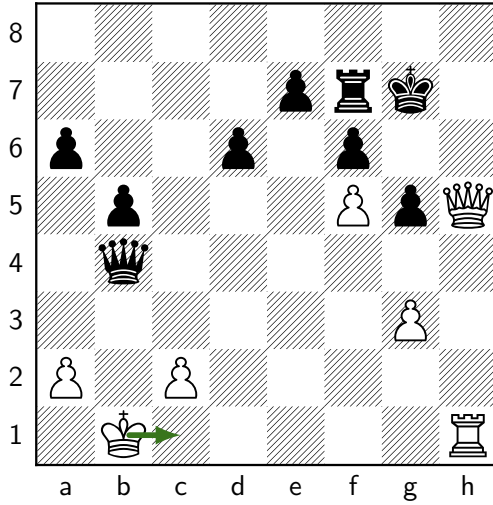
Figure 18: **Move 24, Black: Kc8.** (a) Our position; Melanie plays Kc8 (d8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#3577` (ply 58) [structural distance $d=90$, QAP obj=68]: the master played Kd1; under μ our Kd1 plays the role of that game’s Kc1; cross-type roles: our Ng8 as their Pa2, our Rc7 as their Pc2; transferred move d8c8 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#686` (ply 60) [structural distance $d=81$, QAP obj=68]: the master played Kg2; under μ our Kd1 plays the role of that game’s Kg1; cross-type roles: our Pg7 as their Rc2, our Rc7 as their Ph2; transferred move d8c8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#1985` (ply 61) [structural distance $d=75$, QAP obj=68]: the master played Kb8; under μ our Kd1 plays the role of that game’s Kc1; cross-type roles: our Ng8 as their Pb7, our Qh1 as their Ra5; transferred move d8c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kc8 (summed QAP objective 206, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



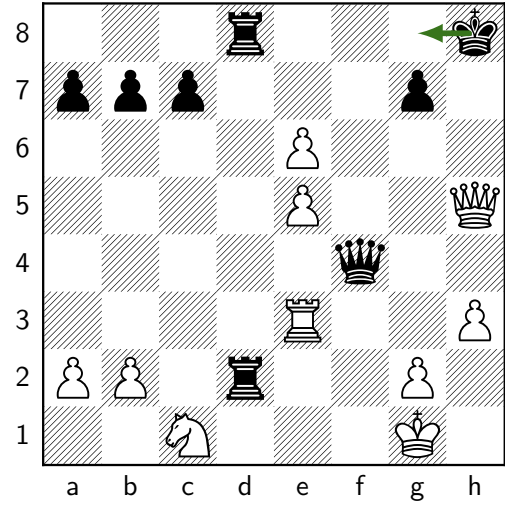
(a)



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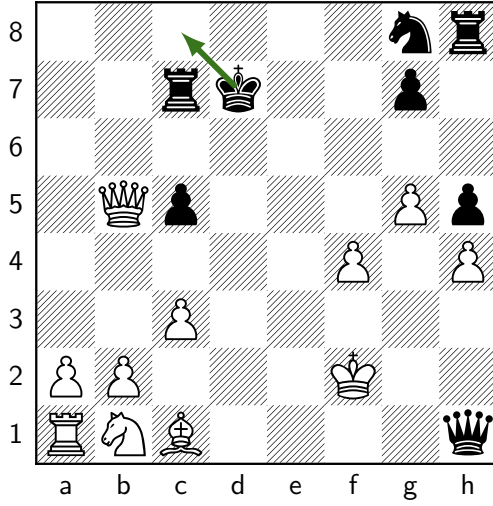


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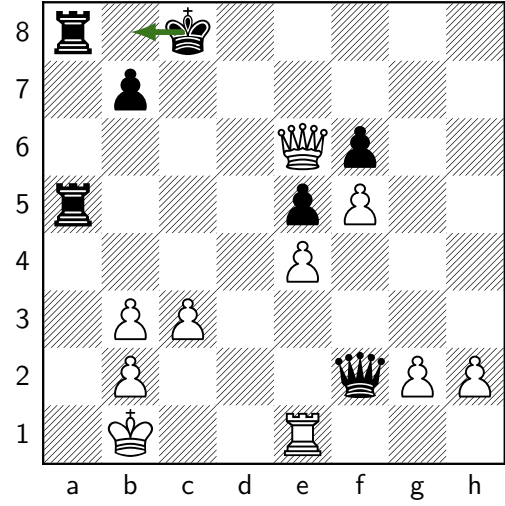


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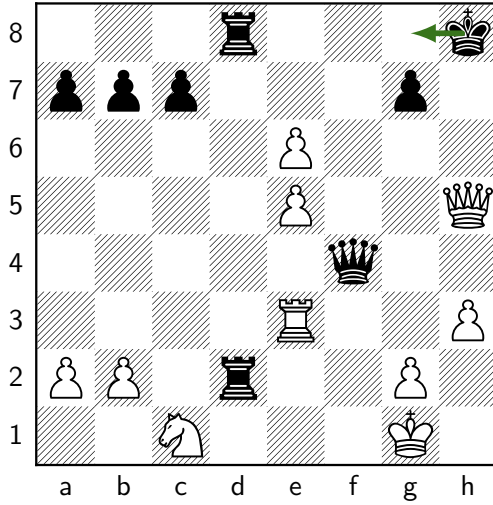
Figure 19: **Move 25, Black: Kd7.** (a) Our position; Melanie plays Kd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1985` (ply 61) [structural distance $d=84$, QAP obj=70]: the master played Kb8; under μ our *Kc1* plays the role of that game’s *Kc1*; cross-type roles: our *Ng8* as their *pf5*, our *ra1* as their *pg2*; transferred move c8d7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3577` (ply 56) [structural distance $d=84$, QAP obj=48]: the master played Kc1; under μ our *Kc1* plays the role of that game’s *Kb1*; cross-type roles: our *Ng8* as their *Pa2*, our *Rh8* as their *Pc2*; transferred move c8d7 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2674` (ply 63) [structural distance $d=82$, QAP obj=47]: the master played Kg8; under μ our *Kc1* plays the role of that game’s *Kh1*; cross-type roles: our *Ng8* as their *Pa7*, our *nb1* as their *Pg7*; transferred move c8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd7 (summed QAP objective 167, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



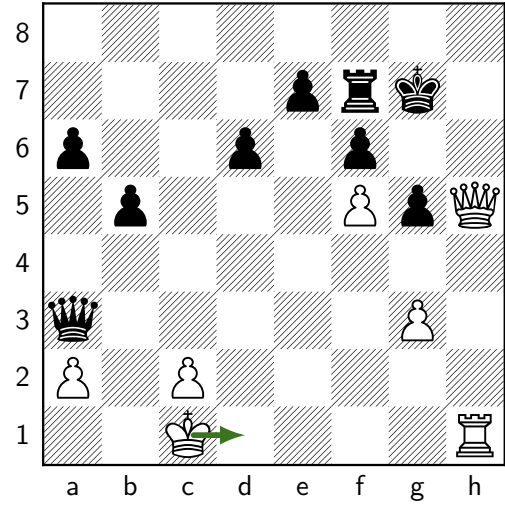
(a)



(b)

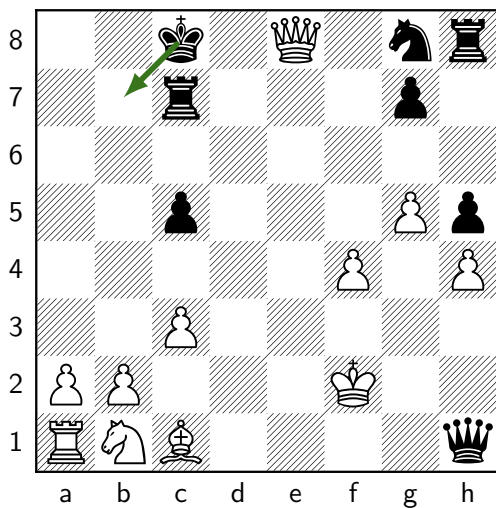


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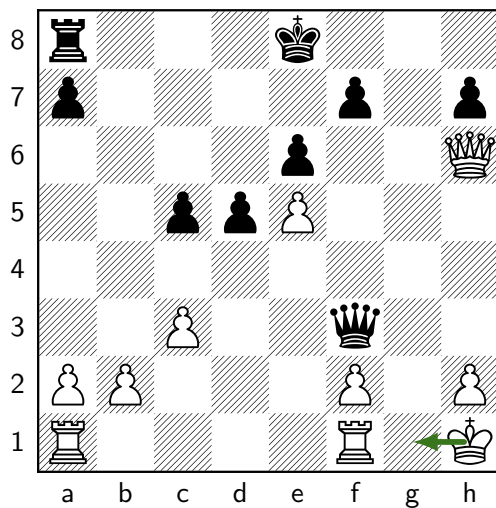


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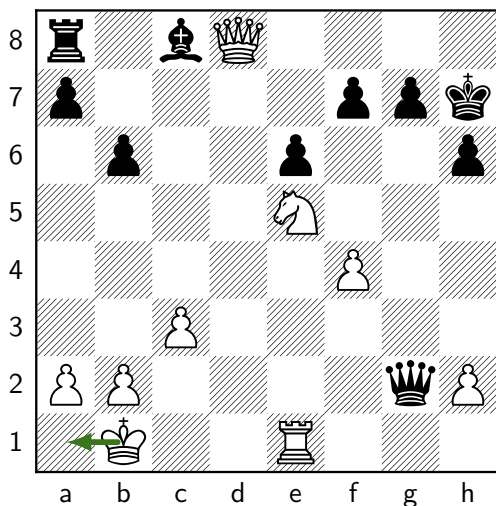
Figure 20: **Move 26, Black: Kc8.** (a) Our position; Melanie plays Kc8 (d7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1985` (ply 61) [structural distance $d=72$, QAP obj=70]: the master played Kb8; under μ our Kd2 plays the role of that game’s Kc1; cross-type roles: our Rh8 as their Pb7, our Qh1 as their Ra5; transferred move d7c8 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2674` (ply 63) [structural distance $d=65$, QAP obj=47]: the master played Kg8; under μ our Kd2 plays the role of that game’s Kh1; cross-type roles: our Ng8 as their Pa7, our Rh8 as their pe6; transferred move d7c8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#3577` (ply 58) [structural distance $d=72$, QAP obj=69]: the master played Kd1; under μ our Kd2 plays the role of that game’s Kc1; cross-type roles: our Rh8 as their pd6, our Ng8 as their pf6; transferred move d7e7. Synthesis: 2 retrieved analogs transferred to Kc8 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



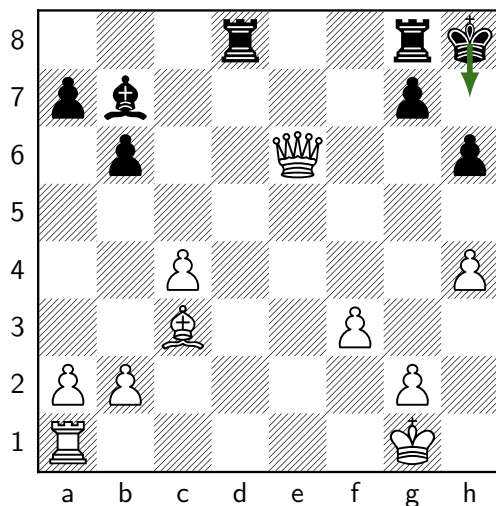
(a)



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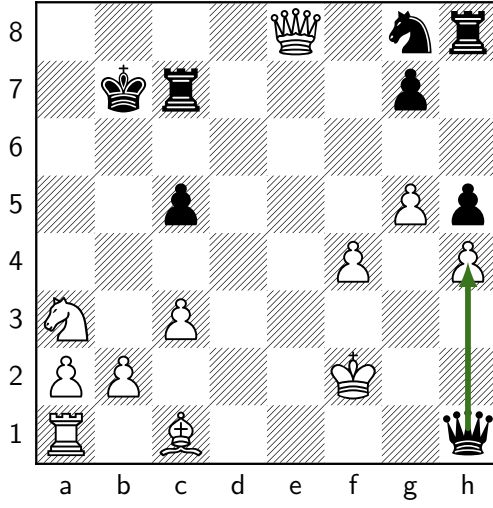


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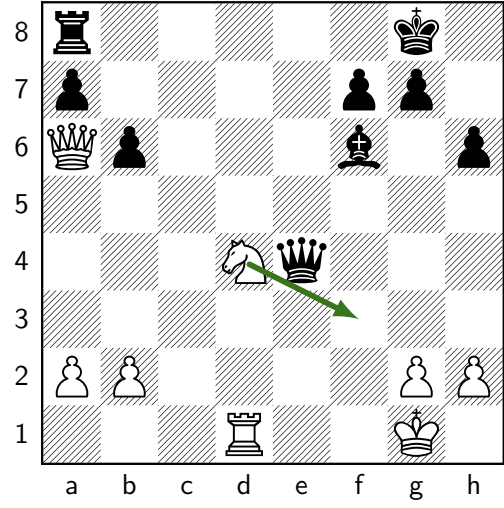


(d)

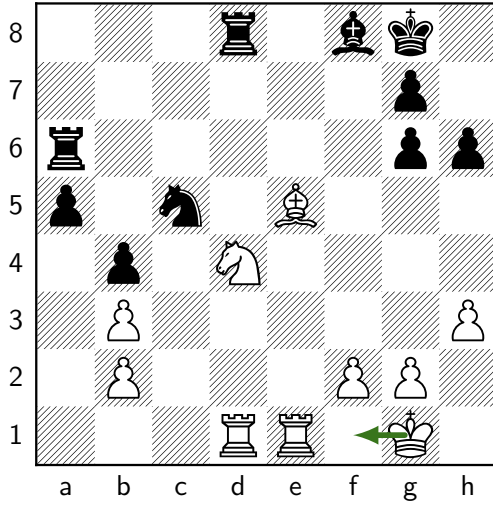
Figure 21: **Move 27, Black: Kb7**. (a) Our position; Melanie plays Kb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#3286` (ply 48) [structural distance $d=75$, QAP obj=84]: the master played Kg1; under μ our Kc1 plays the role of that game’s Kh1; cross-type roles: our bc1 as their Ra1, our Qh1 as their Pa2; transferred move c8b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2225` (ply 48) [structural distance $d=86$, QAP obj=17]: the master played Ka1; under μ our Kc1 plays the role of that game’s Kb1; cross-type roles: our pc3 as their qg2, our ra1 as their Pc3; transferred move c8b7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#3288` (ply 63) [structural distance $d=85$, QAP obj=3]: the master played Kh7; under μ our Kc1 plays the role of that game’s Kh1; cross-type roles: our Ng8 as their Rg8, our Rc7 as their Pa7; transferred move c8b7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kb7 (summed QAP objective 189, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



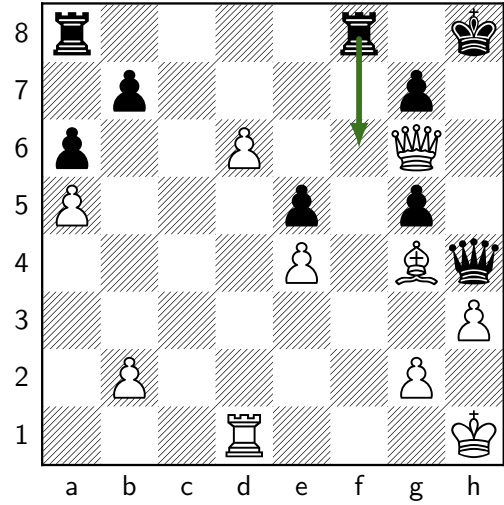
(a)



(b)

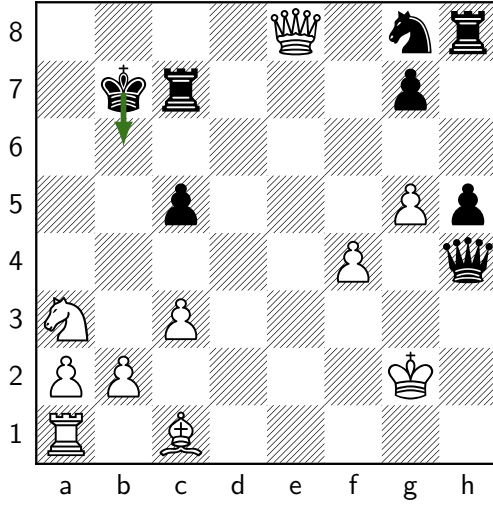


(c)

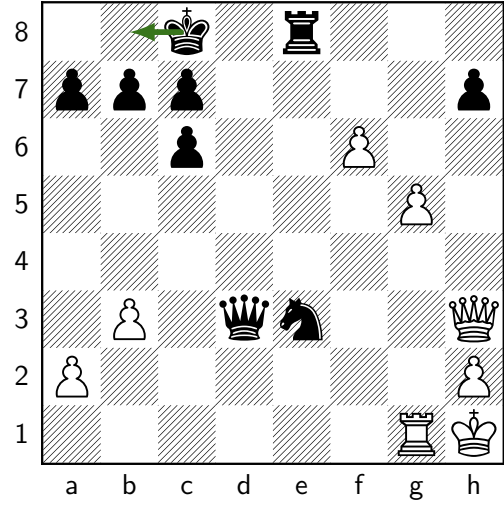


(d)

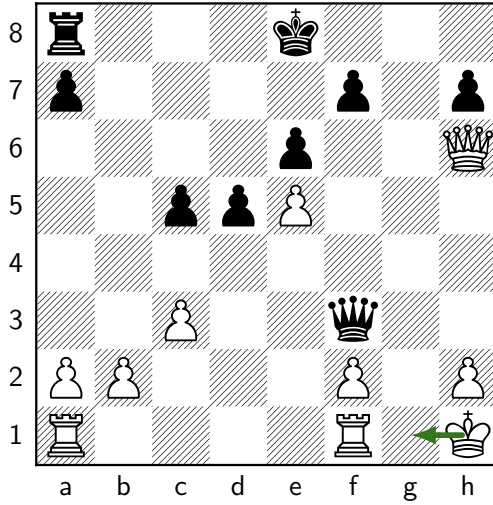
Figure 22: **Move 28, Black: Qxh4+.** (a) Our position; Melanie plays Qxh4+ (h1h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#1015` (ply 48) [structural distance $d=88$, QAP obj=40]: the master played Nf3; under μ our Ng1 plays the role of that game’s Nd4; cross-type roles: our Qh1 as their Pg2, our Ng8 as their pb6; transferred move g8f6. (c) Retrieved from `lichess_elite_2024-09_p0023#3304` (ply 60) [structural distance $d=85$, QAP obj=29]: the master played Kf1; under μ our Kb2 plays the role of that game’s Kg1; cross-type roles: our pg5 as their Re1, our bc1 as their Pb3; transferred move b7b6. (d) Retrieved from `lichess_elite_2024-09_p0025#2183` (ply 65) [structural distance $d=67$, QAP obj=27]: the master played Rf6; under μ our Rc2 plays the role of that game’s Rf1; cross-type roles: our Pc5 as their Ra8, our na3 as their Pa6; transferred move c7e7. Synthesis: no retrieved analog transferred to Qxh4+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



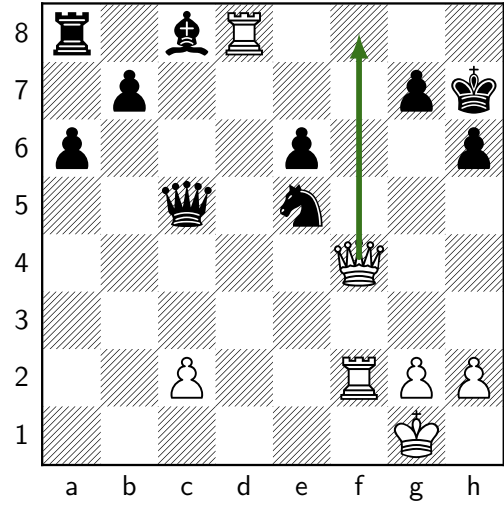
(a)



(b)

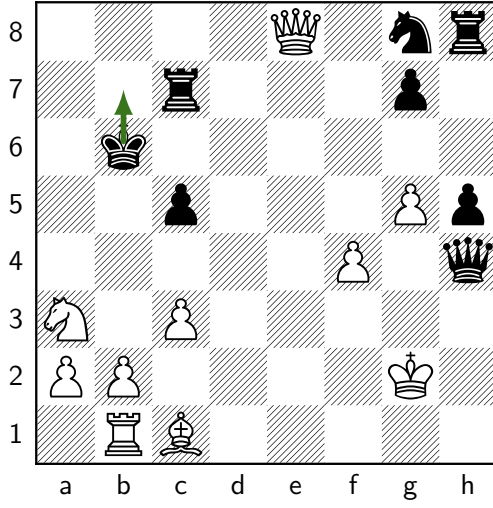


(c)

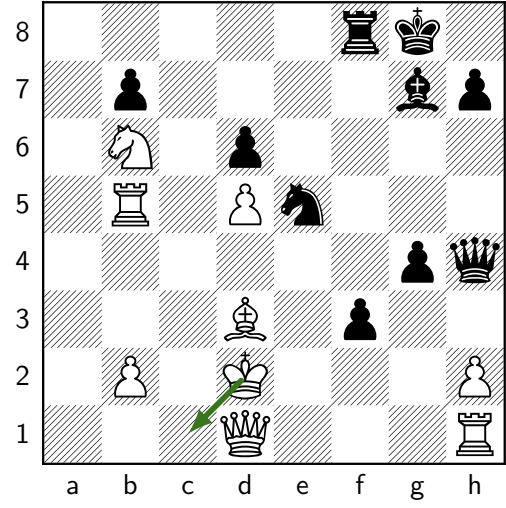


(d)

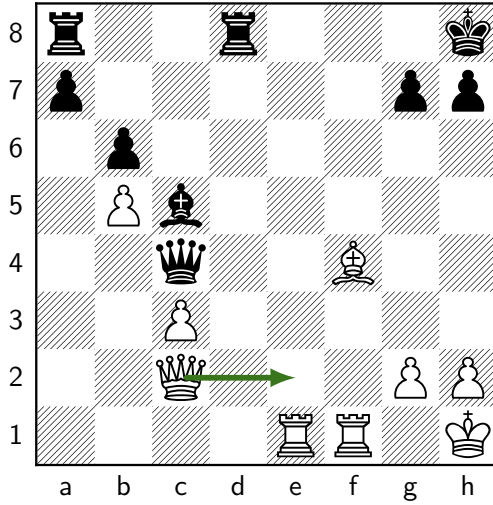
Figure 23: **Move 29, Black: Kb6.** (a) Our position; Melanie plays Kb6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#913` (ply 63) [structural distance $d=80$, QAP obj=20]: the master played Kb8; under μ our Kb2 plays the role of that game’s Kc1; cross-type roles: our na3 as their Pc6, our Qh4 as their pf6; transferred move b7b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3286` (ply 48) [structural distance $d=73$, QAP obj=18]: the master played Kg1; under μ our Kb2 plays the role of that game’s Kh1; cross-type roles: our Ph5 as their Ra1, our Rc7 as their Pa2; transferred move b7b6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#934` (ply 58) [structural distance $d=74$, QAP obj=39]: the master played Qf8; under μ our Qh5 plays the role of that game’s Qf4; cross-type roles: our Ng8 as their Rf2, our Rc7 as their Ph2; transferred move h4e1. Synthesis: 3 retrieved analogs transferred to Kb6 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



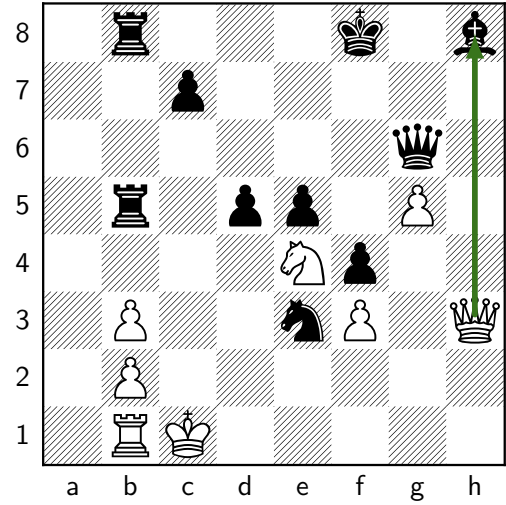
(a)



(b)

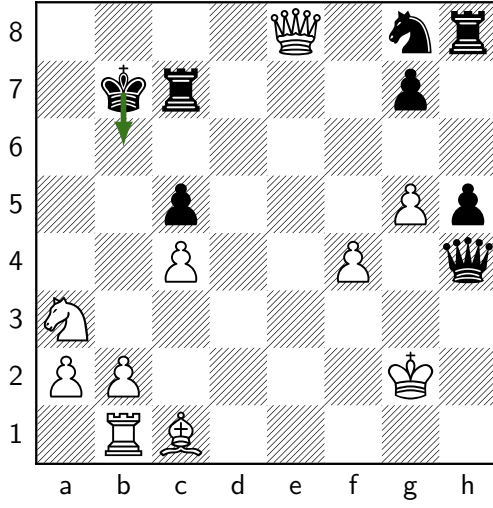


(c)

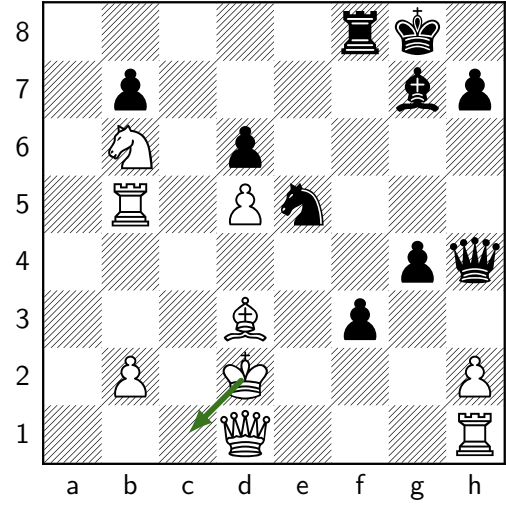


(d)

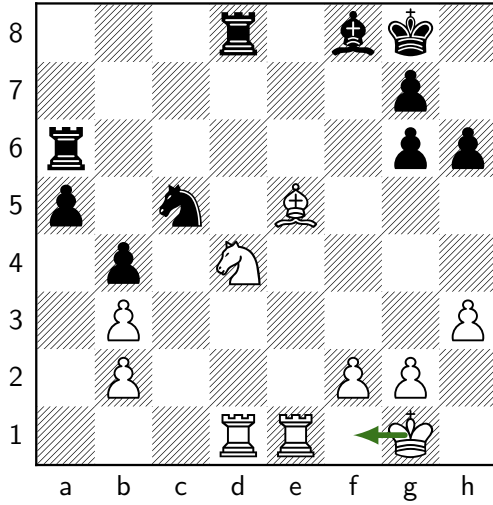
Figure 24: **Move 30, Black: Kb7**. (a) Our position; Melanie plays Kb7 (b6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2814` (ply 60) [structural distance $d=78$, QAP obj=32]: the master played Kc1; under μ our Kb3 plays the role of that game’s Kd2; cross-type roles: our na3 as their Pb2, our Ng8 as their Ph2; transferred move b6b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#1398` (ply 46) [structural distance $d=79$, QAP obj=52]: the master played Qe2; under μ our Rh1 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Re1, our Ng8 as their Rf1; transferred move h8h6. (d) Retrieved from `lichess_elite_2024-09_p0020#2472` (ply 72) [structural distance $d=68$, QAP obj=42]: the master played Qxh8+; under μ our Pc4 plays the role of that game’s Qh3 (a **cross-type** correspondence); cross-type roles: our pa2 as their Rb1, our rb1 as their Pb2; transferred move c5c4. Synthesis: 1 retrieved analog transferred to Kb7 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



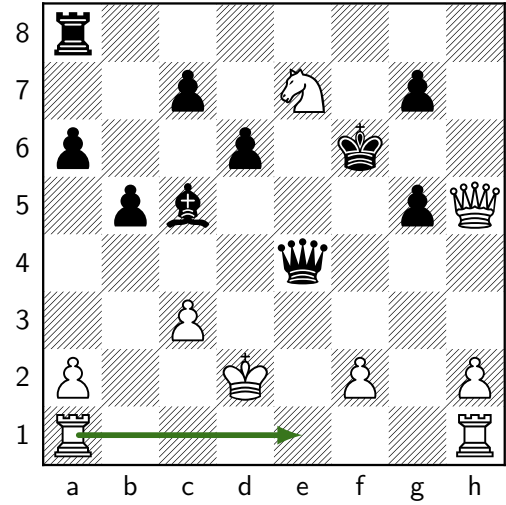
(a)



(b)

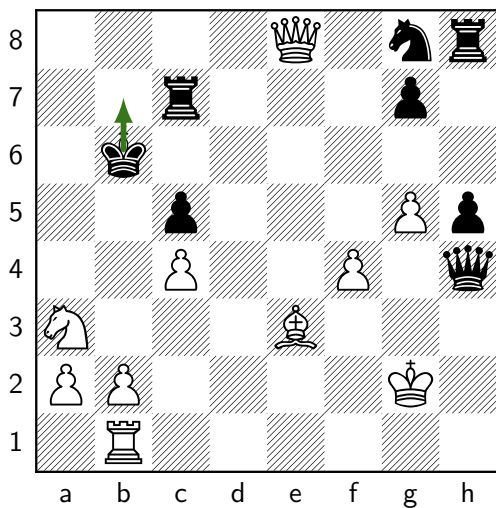


(c)

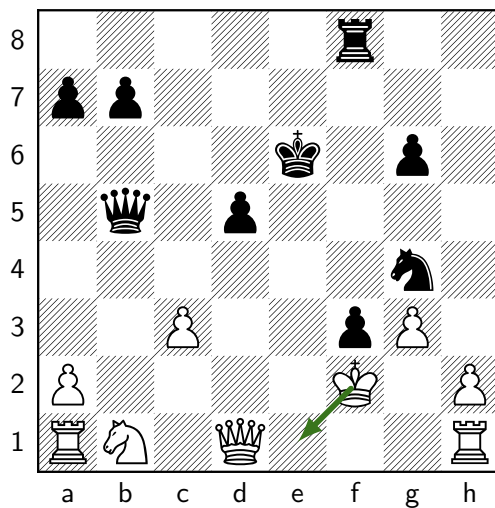


(d)

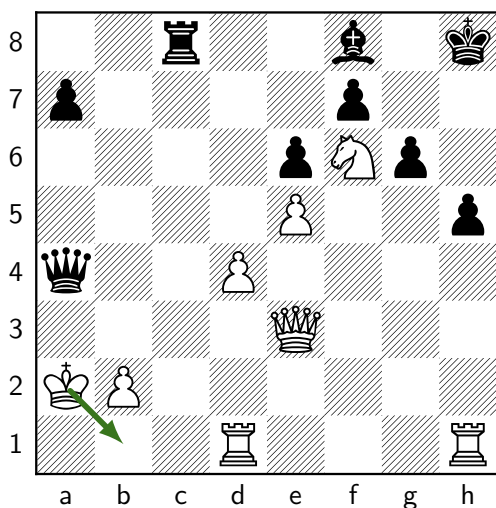
Figure 25: **Move 31, Black: Kb6**. (a) Our position; Melanie plays Kb6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2814` (ply 60) [structural distance $d=66$, QAP obj=31]: the master played Kc1; under μ our Kb2 plays the role of that game’s Kd2; cross-type roles: our Rc7 as their Qd1, our rb1 as their Pb2; transferred move b7b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#3304` (ply 60) [structural distance $d=76$, QAP obj=29]: the master played Kf1; under μ our Kb2 plays the role of that game’s Kg1; cross-type roles: our Pc5 as their Rd1, our Rh8 as their Pf2; transferred move b7b6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#3632` (ply 52) [structural distance $d=68$, QAP obj=30]: the master played Rae1; under μ our Rc2 plays the role of that game’s Ra1; cross-type roles: our pg5 as their Rh1, our rb1 as their Pc3; transferred move c7c8. Synthesis: 2 retrieved analogs transferred to Kb6 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



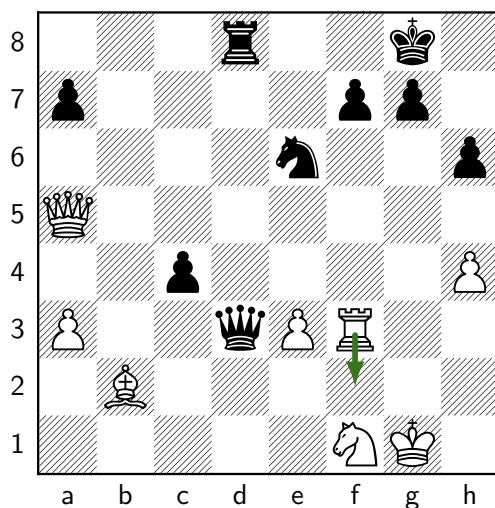
(a)



(b)

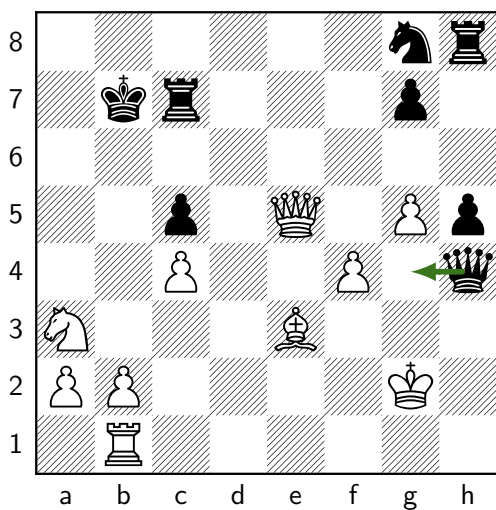


(c)

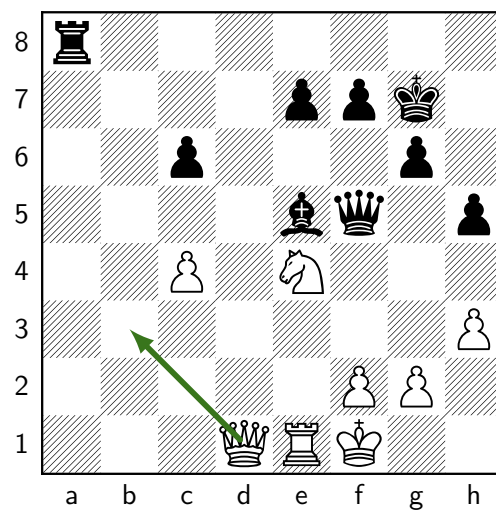


(d)

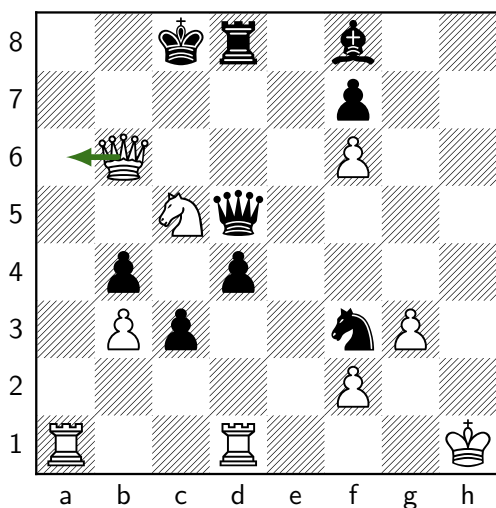
Figure 26: **Move 32, Black: Kb7**. (a) Our position; Melanie plays Kb7 (b6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2617` (ply 52) [structural distance $d=69$, QAP obj=11]: the master played Kc1; under μ our Kb3 plays the role of that game’s Kf2; cross-type roles: our pa2 as their Ra1, our pf4 as their Nb1; transferred move b6b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#3019` (ply 64) [structural distance $d=68$, QAP obj=6]: the master played Kb1; under μ our Kb3 plays the role of that game’s Ka2; cross-type roles: our Qh4 as their Rh1, our Rc7 as their Pb2; transferred move b6b7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#747` (ply 70) [structural distance $d=78$, QAP obj=61]: the master played Rf2; under μ our Rh1 plays the role of that game’s Rf3; cross-type roles: our Pg7 as their Bb2, our Ng8 as their Pe3; transferred move h8h7. Synthesis: 2 retrieved analogs transferred to Kb7 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



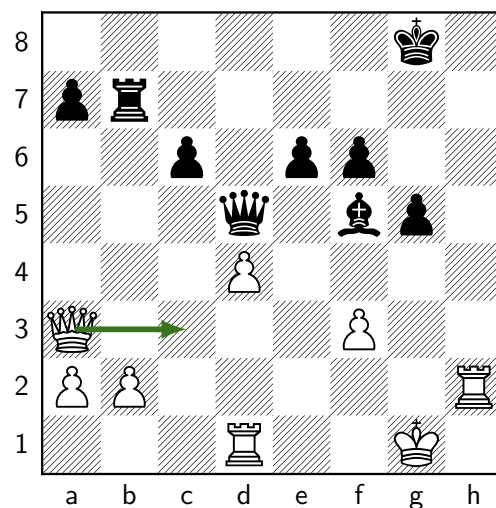
(a)



(b)

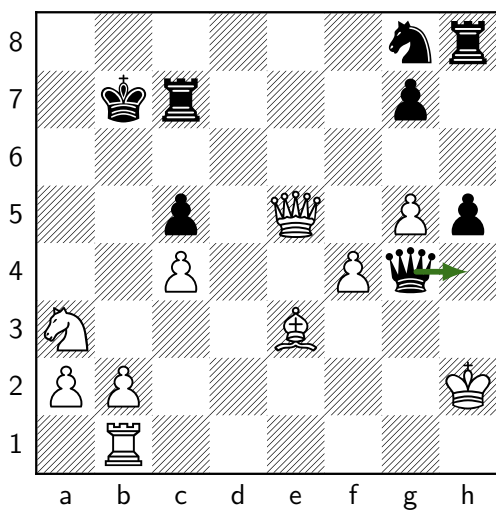


(c)

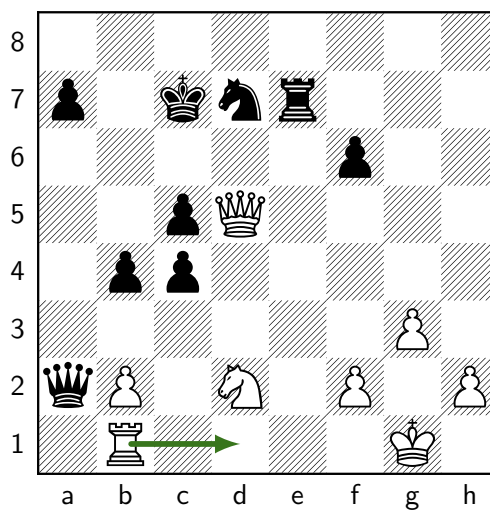


(d)

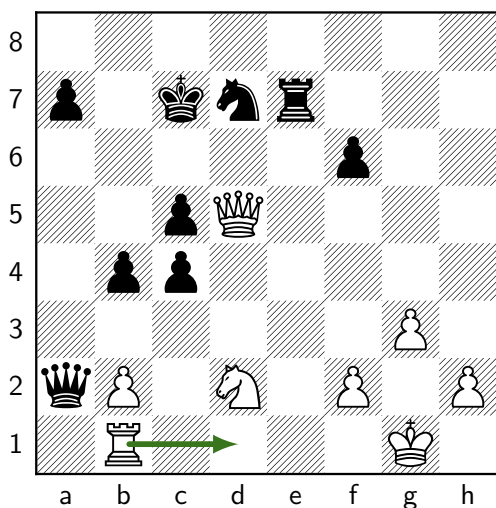
Figure 27: **Move 33, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (h4g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2091` (ply 48) [structural distance $d=82$, QAP obj=57]: the master played Qb3; under μ our Qh5 plays the role of that game’s Qd1; cross-type roles: our Ph5 as their Re1, our Ng8 as their Pg2; transferred move h4g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#652` (ply 70) [structural distance $d=82$, QAP obj=39]: the master played Qa6+; under μ our Qh5 plays the role of that game’s Qb6; cross-type roles: our Ng8 as their Ra1, our qe5 as their Rd1; transferred move h4g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#392` (ply 56) [structural distance $d=80$, QAP obj=27]: the master played Qc3; under μ our Qh5 plays the role of that game’s Qa3; cross-type roles: our pa2 as their Rd1, our Ng8 as their Pb2; transferred move h4g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qg4+ (summed QAP objective 122, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



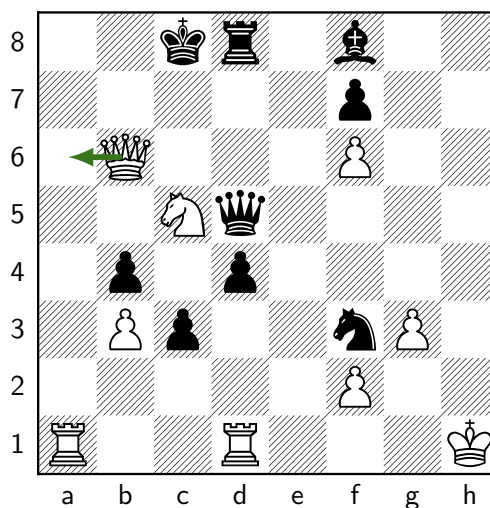
(a)



(b)

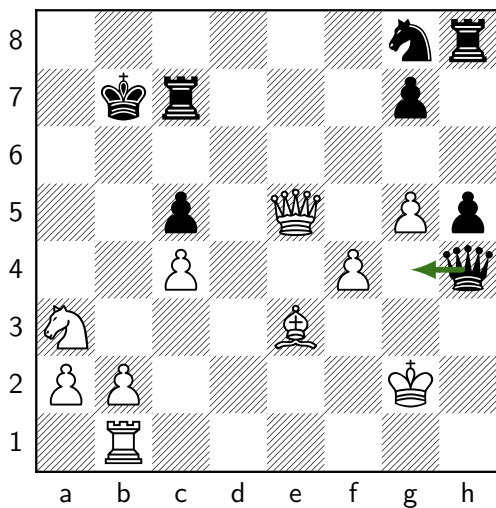


(c)

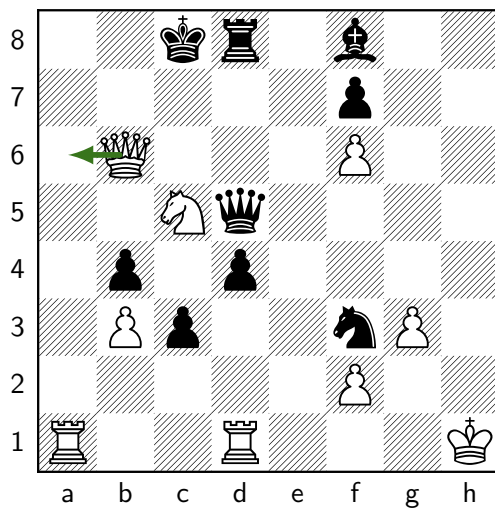


(d)

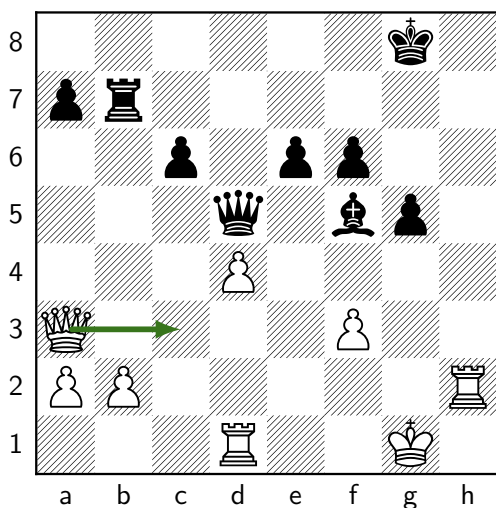
Figure 28: **Move 34, Black: Qh4+.** (a) Our position; Melanie plays Qh4+ (g4h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2324` (ply 52) [structural distance $d=79$, QAP obj=46]: the master played Rd1; under μ our Rc2 plays the role of that game’s Rb1; cross-type roles: our be3 as their Nd2, our na3 as their Pg3; transferred move c7f7. (c) Retrieved from `lichess_elite_2024-09_p0022#1828` (ply 52) [structural distance $d=79$, QAP obj=46]: the master played Rd1; under μ our Rc2 plays the role of that game’s Rb1; cross-type roles: our be3 as their Nd2, our na3 as their Pg3; transferred move c7f7. (d) Retrieved from `lichess_elite_2024-09_p0030#652` (ply 70) [structural distance $d=79$, QAP obj=40]: the master played Qa6+; under μ our Qg5 plays the role of that game’s Qb6; cross-type roles: our pg5 as their Ra1, our ge5 as their Rd1; transferred move g4d7. Synthesis: no retrieved analog transferred to Qh4+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



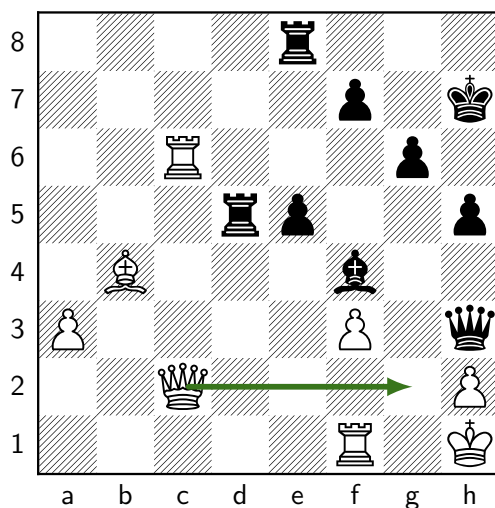
(a)



(b)

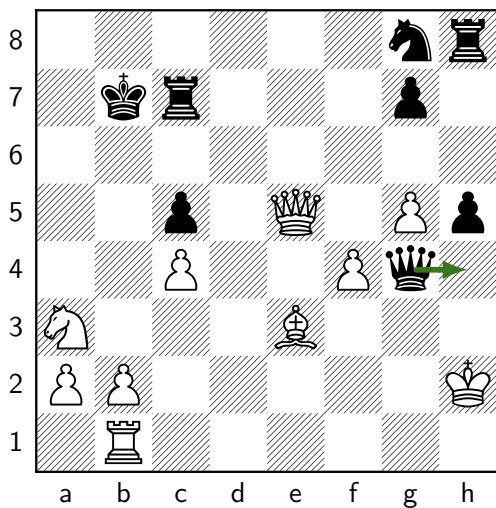


(c)

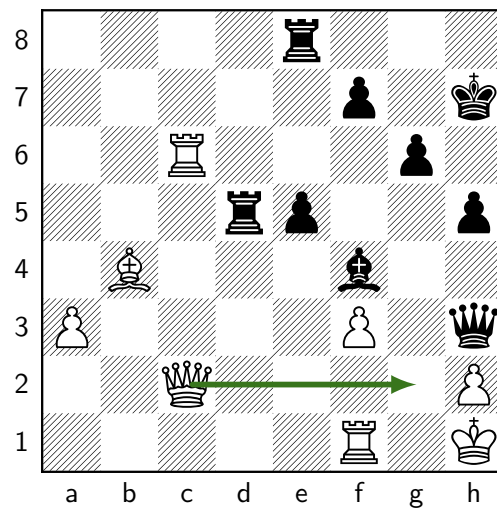


(d)

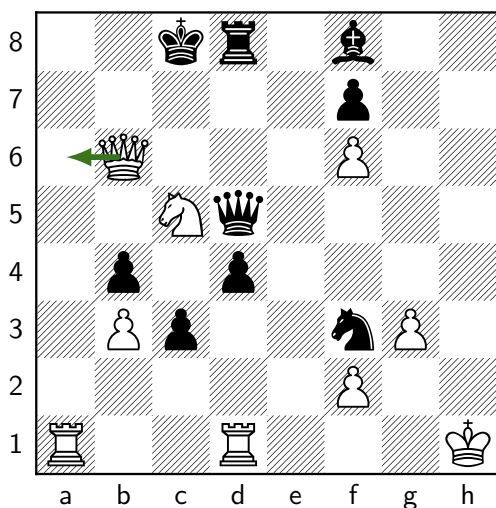
Figure 29: **Move 35, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (h4g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#652` (ply 70) [structural distance $d=82$, QAP obj=39]: the master played Qa6+; under μ our Qh5 plays the role of that game’s Qb6; cross-type roles: our Ng8 as their Ra1, our qe5 as their Rd1; transferred move h4g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#392` (ply 56) [structural distance $d=80$, QAP obj=27]: the master played Qc3; under μ our Qh5 plays the role of that game’s Qa3; cross-type roles: our pa2 as their Rd1, our Ng8 as their Pb2; transferred move h4g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#2507` (ply 80) [structural distance $d=84$, QAP obj=55]: the master played Qg2; under μ our Qh5 plays the role of that game’s Qc2; cross-type roles: our na3 as their Bb4, our pb2 as their rd5; transferred move h4g5. Synthesis: 2 retrieved analogs transferred to Qg4+ (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



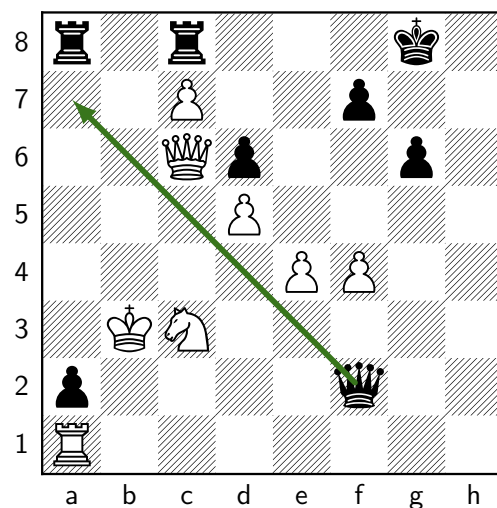
(a)



(b)



(c)



(d)

Figure 30: **Move 36, Black: Qh4+.** (a) Our position; Melanie plays Qh4+ (g4h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#2507` (ply 80) [structural distance $d=83$, QAP obj=55]: the master played Qg2; under μ our Qg5 plays the role of that game’s Qc2; cross-type roles: our Ng8 as their Bb4, our Ph5 as their rd5; transferred move g4c8. (c) Retrieved from `lichess_elite_2024-09_p0030#652` (ply 70) [structural distance $d=79$, QAP obj=40]: the master played Qa6+; under μ our Qg5 plays the role of that game’s Qb6; cross-type roles: our pg5 as their Ra1, our qe5 as their Rd1; transferred move g4d7. (d) Retrieved from `lichess_elite_2024-09_p0026#1459` (ply 83) [structural distance $d=77$, QAP obj=39]: the master played Qa7; under μ our Rh1 plays the role of that game’s Qf7 (a **cross-type** correspondence); cross-type roles: our Qg4 as their Rc8, our Ng8 as their Pf7; transferred move h8h6. Synthesis: no retrieved analog transferred to Qh4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.